

ABBREVIATIONS

ABV	ABOVE	LAT	LEAVING AIR TEMPERATURE
AD	ACCESS DOOR	LBS	POUNDS
AHU	AIR HANDLING UNIT	LF	LINEAR FOOT/FEET
AL	ACOUSTIC LINED	LWG	LEAVING
AP	ACCESS PANEL	LWR	LOW WALL REGISTER
APD	AIR PRESSURE DROP	LWT	LEAVING WATER TEMPERATURE
ARCH	ARCHITECT/ARCHITECTURAL	MAU	MAKEUP AIR UNIT
ARV	AUTOMATIC RELIEF VALVE or ACID RESISTANT WASTE	MAX	MAXIMUM
ARW		MBH	1000 BRITISH THERMAL UNITS PER HOUR
BDD	BACKDRAFT DAMPER	MCA	MAXIMUM CIRCULATING CAPACITY
BFP	BACKFLOW PREVENTER	MCC	MOTOR CONTROL CENTER
BHP	BRAKE HORSEPOWER	MECH	MECHANICAL
BG	BELOW GROUND	MFR	MANUFACTURER
BTU	BRITISH THERMAL UNIT	MIN	MINIMUM
BTUH	BRITISH THERMAL UNITS/HOUR	MISC	MISCELLANEOUS
C	CENTIGRADE	MOP	MAXIMUM OVERCURRENT PROTECTION
CC	COOLING COIL	MUA	MAKE UP AIR
CD	CEILING DIFFUSER	MTG	MOUNTING
CFM	CUBIC FEET PER MINUTE	N/A	NOT APPLICABLE
CG	CEILING GRILLE	NC	NORMALLY CLOSED
CLG	CEILING	NO	NORMALLY OPEN
CMU	CONCRETE MASONRY UNIT	NC	NOISE CRITERIA
CM	CONDENSATE METER	N.I.C.	NOT IN CONTRACT
CO	CLEANOUT	NPT	NATIONAL PIPE THREAD
CO2	CARBON DIOXIDE	NTS	NOT TO SCALE
CONC	CONCRETE	OPD	OPPOSED BLADE DAMPER
CONT	CONTINUATION	OC	ON CENTER
CUH	CUPPER	OD	OUTSIDE DIAMETER
CUH	CABINET UNIT HEATER	OPNG	OPENING
DB	DECEIBLE or DRY BULB DIAMETER or PHASE	OW	ONLY WASTE
DDC	DIRECT DIGITAL CONTROL		
DIA	DIAMETER		
DM	DIMENSION		
DN	DOWN		
DOAS	DEDICATED OUTSIDE AIR SYSTEM		
DPS	DIFFERENTIAL PRESSURE SWITCH		
DPR	DAMPEN		
DWG	DRAWING		
EA	EXHAUST AIR		
EAT	ENTERING AIR TEMPERATURE		
EF	EXHAUST FAN		
EG	EXHAUST GRILLE		
ELEC	ELECTRIC or ELECTRICAL		
ELEV	ELEVATION		
EMCS	ENERGY MANAGEMENT CONTROL SYSTEM		
ESP	EXTERNAL STATIC PRESSURE		
EWT	ENTERING WATER TEMPERATURE		
EXH	EXHAUST		
EXIST	EXISTING		
(E)	EXISTING		
FA	FAHRENHEIT		
FA	FACE AREA or FREE AREA		
FCO	FLOOR CLEANOUT		
FCU	FAN COIL UNIT		
FD	FLOOR DRAIN or FIRE DAMPER		
FFD	FUNNEL FLOOR DRAIN		
FF	FINAL FILTER		
FLR	FLOOR		
FP	FIRE PUMP		
FPC	FIRE PUMP CONTROLLER		
FPM	FEET PER MINUTE		
FPS	FEET PER SECOND		
FSD	FIRE/SMOKE DAMPER		
FT	FOOT or FEET		
FV	FACE VELOCITY		
GA	GAGE or GAUGE		
GAL	GALLON		
GALV	GALVANIZED		
GPH	GALLONS PER HOUR		
GPM	GALLONS PER MINUTE		
HC	HEATING COIL		
HP	HORSE POWER		
HR	HOUR		
HTG	HEATING		
HVAC	HEATING, VENTILATION AND AIR CONDITIONING		
HWG	HIGH WALL GRILLE		
HWR	HIGH WALL REGISTER		
HZ	HERTZ		
I.D.	INSIDE DIAMETER		
I.E.	INVERT ELEVATION		
IN	INCH or INCHES		
KW	KILOWATT		
KWH	KILOWATT HOUR		

DESIGN CONDITIONS

LOCATION	SPOKANE, WA
OUTDOORS	SUMMER 92.9°F DB / 82.8°F WB WINTER 5.1°F DB EVAPORATION 65.2°F WB/ 86.9°F MCOB
INDOOR	COOLING 75°F HEATING 70°F
VENTILATION	PER 2018 WSCM
ELEVATION	1890 FT ABOVE SEA LEVEL

HVAC FITTINGS & SYMBOLS

(PERMITTED DUCT FITTINGS)

BRANCH DUCT TAKEOFFS

RECTANGULAR DUCT
CONICAL SPIN-IN
ROUND DUCT RUNOUT TO SINGLE DIFFUSER OR GRILLE ONLY

ONLY PERMITTED FOR AIR VELOCITY UNDER 1,500 FPM

ROUND OR RECTANGULAR

R=1/2W FOR DUCTS W/ AIR VELOCITY UNDER 1,500 FPM

R=W FOR DUCTS W/ AIR VELOCITY OVER 1,500 FPM AND ALL DUCTS UPSTREAM OF TERMINAL UNITS

CHANGE OF DUCT SIZE

TRANSITION
END TAP

A ≤ 15" FOR DUCTS W/ AIR VELOCITY OVER 1,500 FPM

A ≤ 30" FOR DUCTS W/ AIR VELOCITY UNDER 1,500 FPM

CONICAL SPIN-IN OR TRANSITION TO AT LEAST 70% GREATER THAN AREA OF SMALLER DUCT. SEE MAX TRANSITION ANGLE.

OFFSETS

NOTE: OFFSETS (TO AVOID OBSTRUCTIONS) ARE REQUIRED BUT ARE NOT NECESSARILY SHOWN ON THE PLANS.

45° OR 90° RADIUS ELBOWS

A ≤ 15" FOR DUCTS W/ AIR VELOCITY OVER 1,500 FPM

A ≤ 30" FOR DUCTS W/ AIR VELOCITY UNDER 1,500 FPM

(OBTAIN PRIOR APPROVAL FOR ALL RETURN & EXHAUST LOCATIONS)

TURNING VANES

ACCESS PANEL (TYPICAL) REQUIRED FOR RETURN & EXHAUST DUCTS

DUCT ELBOWS

R=1/2W FOR DUCTS W/ AIR VELOCITY UNDER 1,500 FPM

R=W FOR DUCTS W/ AIR VELOCITY OVER 1,500 FPM AND ALL DUCTS UPSTREAM OF TERMINAL UNITS

12"x12" ACCESS PANEL ON BOTTOM OF DUCT UPSTREAM OF TURNING VANES. REQUIRED FOR RETURN AND EXHAUST DUCTS

TURNING VANES

OTHER HVAC SYMBOLS

AIR FLOW OUT
AIR FLOW IN
THERMOSTAT
HUMIDISTAT
SWITCH
CO2 SENSOR
CO SENSOR
NOX SENSOR

BASEBOARD IDENTIFICATION
DIFFUSER IDENTIFICATION
VAV IDENTIFICATION
HEATING COIL

SUPPLY AIR (S/A) TERMINAL
RETURN AIR (R/A) TERMINAL
EXHAUST AIR (E/A) TERMINAL
LINEAR SLOT DIFFUSER
FLEX DUCT OR FLEX PIPE
UNIT HEATER (HORIZONTAL)
BASEBOARD W/ COVER

DUCTWORK

DUCT SIZE IN INCHES, FIRST SIZE IS SIZE SHOWN

18a ROUND
18x12 RECTANGLE
18/12 OVAL

INSULATED DUCT

ACOUSTIC LINED DUCT

DUCT RISE IN DIRECTION OF ARROW

DUCT DROP IN DIRECTION OF ARROW

ROUND DUCT IN INCHES

CHANGE IN DUCT SIZE AND/OR SHAPE

VOLUME DAMPER

FIRE DAMPER

SMOKE DAMPER

FIRE/SMOKE DAMPER

MOTORIZED DAMPER

DUCT RISE/DROP/SECTION

SUPPLY AIR (S/A) - CROSS

RECTANGULAR DUCT TURNED UP/SECTION
ROUND DUCT TURNED UP/SECTION
RECTANGULAR DUCT TURNED DOWN/HIDDEN SECTION
ROUND DUCT TURNED DOWN/HIDDEN SECTION

RETURN AIR (R/A) - SLASH

RECTANGULAR DUCT TURNED UP/SECTION
ROUND DUCT TURNED UP/SECTION
RECTANGULAR DUCT TURNED DOWN/HIDDEN SECTION
ROUND DUCT TURNED DOWN/HIDDEN SECTION

EXHAUST AIR (E/A) - WYE
OUTSIDE AIR (O/A) - REVERSE WYE

RECTANGULAR DUCT TURNED UP/SECTION
ROUND DUCT TURNED UP/SECTION
RECTANGULAR DUCT TURNED DOWN/HIDDEN SECTION
ROUND DUCT TURNED DOWN/HIDDEN SECTION

ALL SYMBOLS AND ABBREVIATIONS DO NOT NECESSARILY APPEAR ON DRAWINGS

CONTRACTOR PHASING NOTES

- CONTRACTOR TO SUBMIT CONSTRUCTION PHASING PLAN TO STA AND STA MUST ACCEPT PHASING PLAN PRIOR TO START OF WORK.
- CONSTRUCTION PHASING TO PLACE PRIORITY ON BOILER #1 REPLACEMENT AND BOILER SHOULD BE REPLACED BEFORE OTHER EQUIPMENT REPLACEMENT.
- PLAZA HEATING SYSTEM, BOILER #2 AND AT LEAST ONE OF THE HEATING WATER PUMPS TO BE REPLACED SHOULD REMAIN OPERATIONAL AT ALL TIMES DURING CONSTRUCTION.
- HEATING WATER PUMP REPLACEMENT SHOULD BE PHASED SO THAT ONE PUMP REMAINS OPERATIONAL AT ALL TIMES.
- COOLING TOWER DEMOLITION AND NEW TOWER CONSTRUCTION TO BE PHASED AS NOT TO DISRUPT STA PLAZA, STREET AND SIDEWALK OPERATION.
- COOLING TOWER, PUMP AND BOILER REPLACEMENT, CRANE OPERATIONS AND CONSTRUCTION LAYDOWN IS NOT TO DISRUPT STA PLAZA OPERATIONS.

PIPING LEGEND

GATE VALVE
NEEDLE VALVE
BALL VALVE
GLOBE VALVE
PLUG VALVE
STOP COCK
OS & Y GATE VALVE
AUTOMATIC CONTROL VALVE (PNEUMATIC)
3-WAY AUTO. CONTROL VALVE (PNEUMATIC)
MOTORIZED (M) OR SOLENOID (S) VALVE
3-WAY MOTORIZED (M) OR SOLENOID (S) VALVE
BUTTERFLY VALVE
FUSIBLE OIL SAFETY VALVE
CHECK VALVE
SPRING CHECK VALVE
BACKFLOW PREVENTER
BALANCE VALVE
BALANCE VALVE (AUTOMATIC)
PRESSURE REDUCING VALVE
PRESSURE RELIEF VALVE
T&P RELIEF VALVE
BASKET STRAINER
STRAINER
STRAINER WITH BLOW DOWN (1)
HOSE BIBB
HOSE THREAD DRAIN VALVE (1)
HOSE THREAD DRAIN BALL VALVE (1)
(1) INDICATES TO PROVIDE W/ END CAP.
HOSE QUICK DISCONNECT
BLIND FLANGE
FLANGES
FIRE DEPT. CONNECTION
UNION
REDUCER, CONCENTRIC
REDUCER, ECCENTRIC
PT PLUG
THERMOWELL
THERMOMETER (LIQUID IN GLASS)
DIAL THERMOMETER
PRESSURE GAUGE W/ COCK
PRESSURE GAUGE W/ COCK & SIPHON
AIR VENT
C - COIN
M - MANUAL
A - AUTOMATIC
H - HIGH VOLUME

S=2%

PIPE SLOPE DIRECTION DOWN
FLOW DIRECTION
EXPANSION LOOP
EXPANSION JOINT
EXPANSION GUIDE
ANCHOR
FLEX COUPLING
PUMP (ARROWHEAD INDICATES FLOW DIRECTION)
FLOW METER
PIPE UP
PIPE DOWN
TEE UP
TEE DOWN
ELBOW
TEE
CAP
WALL CLEANOUT
FLOOR CLEAN OUT
YARD CLEAN OUT
FLOOR SINK
FLOOR DRAIN
ROOF OVERFLOW DRAIN
ROOF DRAIN
WATER HAMMER ARRESTOR (PLAN)
WATER HAMMER ARRESTOR (ELEV.)
EMERGENCY EYEWASH/SHOWER
EMERGENCY EYEWASH
SHOWER
CW - COLD WATER MAKE-UP
CDR - CONDENSER WATER RETURN
CDS - CONDENSER WATER SUPPLY
CWR - CHILLED WATER RETURN
CWS - CHILLED WATER SUPPLY
D - DRAIN
HWR - HEATING WATER RETURN
HWS - HEATING WATER SUPPLY
HPR - HEAT PUMP LOOP RETURN
HPS - HEAT PUMP LOOP SUPPLY
G - GAS

GENERAL

NOTES:

- SPECIFIC TO LOCATION INDICATED.
- NOTE APPLIES TO ENTIRE SHEET.
- DESIGNATOR HAS SCHEDULED VALUES. SEE MECHANICAL SCHEDULES.
- PLUMBING FIXTURE TAG
- EXISTING
- NEW
- POINT OF CONNECTION (POC)

LINETYPES

- EXISTING (THIN)
NEW (MEDIUM)
EXISTING TO BE DEMOLISHED
- T - THERMOSTAT, ADJACENT NUMBER INDICATES TERMINAL UNIT CONTROLLED
H - HUMIDISTAT
S - SENSOR
CO2 - CO2 SENSOR
SD - DUCT SMOKE DETECTOR
SP - DUCT STATIC PRESSURE

GENERAL NOTES

- CONTRACTOR SHALL SECURE, MAINTAIN, AND PAY FOR ALL REQUIRED PERMITS, LICENSES, AND INSPECTIONS FOR DURATION OF WORK UNLESS DIRECTED OTHERWISE.
- ALL WORK SHALL COMPLY WITH THE OWNER'S REQUIREMENTS, AND WITH ALL APPLICABLE STATE AND LOCAL CODES, OR AUTHORITY HAVING JURISDICTION.
- SUBSTITUTIONS OF EQUIPMENT OTHER THAN AS SPECIFIED SHALL BE THE COMPLETE RESPONSIBILITY OF THE CONTRACTOR, ANY ADDITIONAL ELECTRICAL, STRUCTURAL, MECHANICAL, OR ARCHITECTURAL REQUIREMENTS SHALL BE PROVIDED AT NO ADDITIONAL EXPENSE TO THE OWNER.
- DRAWINGS ARE DIAGRAMMATIC IN NATURE AND DO NOT ATTEMPT TO SHOW ALL REQUIRED OFFSETS. COORDINATE WITH OTHER TRADES AND PROVIDE ALL NECESSARY OFFSETS. REFER TO ARCHITECTURAL AND STRUCTURAL DRAWINGS FOR CONSTRUCTION DETAILS.
- REFER TO PROJECT MANUAL (SPECIFICATIONS) FOR ADDITIONAL REQUIREMENTS. PLANS AND SPECIFICATIONS SHALL BE TAKEN TOGETHER. PROVIDE ALL WORK CALLED FOR IN EITHER. IN CASE OF CONFLICT BETWEEN SPECIFICATIONS AND PLANS THE MORE STRINGENT SHALL APPLY.
- ALL EQUIPMENT AND MATERIAL ON THE JOB SITE PRIOR TO INSTALLATION SHALL BE COVERED AND PROTECTED FROM RAIN, DIRT, DUST, AND DAMAGE.
- VERIFY PHYSICAL DIMENSIONS OF EQUIPMENT TO ENSURE THAT ACCESS CLEARANCES CAN BE MET. COORDINATE FINAL EQUIPMENT LOCATIONS W/ GENERAL CONTRACTOR AND OTHER TRADES TO ALLOW FOR REQUIRED MECHANICAL EQUIPMENT CLEARANCES.
- PROVIDE NEC CODE MINIMUM HORIZONTAL AND VERTICAL WORKING CLEARANCES FOR ALL ELECTRICAL PANELS AND EQUIPMENT. OFFSET MECHANICAL WORK AS REQUIRED.
- CONTRACTOR IS RESPONSIBLE FOR VERIFICATION OF INSTALLATION STANDARDS AND CONSTRUCTION CONDITIONS. CONTRACTOR SHALL FIELD VERIFY ALL DIMENSIONS AND EXISTING CONDITIONS PRIOR TO SHOP FABRICATION AND/OR FIELD INSTALLATION. DISCREPANCIES BETWEEN EXISTING CONDITIONS AND THE CONSTRUCTION DRAWINGS SHALL BE CALLED TO THE ATTENTION OF THE ENGINEER.
- ALL MECHANICAL EQUIPMENT, MATERIALS, AND INSTALLATION SHALL BE PROVIDED BY THE CONTRACTOR. ALL EQUIPMENT SHALL BE COMPLETE, INSTALLED, AND FULLY FUNCTIONAL PRIOR TO FINAL ACCEPTANCE OF THE WORK.
- INSTALL ALL EQUIPMENT PER MANUFACTURERS RECOMMENDATIONS AND REQUIREMENTS. ANY CONFLICTS BETWEEN THE MANUFACTURERS DOCUMENTATION AND THE CONTRACT DOCUMENTS SHALL BE BROUGHT TO THE ATTENTION OF THE ENGINEER.
- DO NOT ALLOW ANY WORK TO BE COVERED UP OR ENCLOSED UNTIL INSPECTED, TESTED AND APPROVED BY OWNER'S REPRESENTATIVE OR AUTHORITY HAVING JURISDICTION.
- MECHANICAL EQUIPMENT SHALL NOT BE USED FOR TEMPORARY HEAT DURING CONSTRUCTION.

GENERAL HVAC NOTES

- CONTRACTOR TO PROVIDE TRANSITION TO CONNECT TO ALL MECHANICAL EQUIPMENT. PROVIDE FLEXIBLE CONNECTION ON INTAKE AND DISCHARGE OF ALL MECHANICAL EQUIPMENT.
- DRAWINGS ARE DIAGRAMMATIC IN NATURE AND DO NOT ATTEMPT TO SHOW ALL REQUIRED OFFSETS. COORDINATE WITH OTHER TRADES AND PROVIDE ALL NECESSARY OFFSETS.
- 24"x24" ACCESS DOORS SHALL BE PROVIDED FOR ALL MECHANICAL EQUIPMENT LOCATED ABOVE GYPSUM BOARD CEILINGS. ALL FINAL ACCESS DOOR LOCATIONS SHALL BE APPROVED BY ARCHITECT PRIOR TO INSTALLATION.

2018 WSEC GENERAL NOTES

- ALL MOTORS SIZED FROM 1/12 HP TO 1 HP SHALL BE ELECTRONICALLY COMMUTATED (EC) MOTORS OR SHALL HAVE A MINIMUM EFFICIENCY OF 70 PERCENT (IN ACCORDANCE WITH DOE 10 CFR 431). MOTORS SHALL ALSO BE EQUIPPED WITH MEANS TO ADJUST MOTOR SPEED PER 2018 WSEC C405.8.
- IN ACCORDANCE WITH C405.8 OF THE 2018 WSEC ALL ELECTRIC MOTORS SHALL MEET THE MINIMUM EFFICIENCY REQUIREMENTS OF TABLES C405.8(1) THROUGH C405.8(4).
- FAN MOTORS SHALL BE THE SMALLEST AVAILABLE MOTOR SIZE GREATER THAN FAN BHP UNLESS OTHERWISE NOTED PER 2018 WSEC C403.8.2.
- ALL MOTORS SHALL BE ENERGY EFFICIENT MOTORS MEETING NEMA STANDARD MG-1 EXCEPT THOSE INCLUDED IN PACKAGED EQUIPMENT WITH EFFICIENCY RATED AS A WHOLE UNIT.
- HEATING AND COOLING LOADS HAVE BEEN PERFORMED PER ASHRAE STANDARDS.
- HVAC EQUIPMENT SIZED NO LARGER THAN THE NEXT LARGEST UNIT AVAILABLE ABOVE THE CALCULATED HEATING/COOLING LOADS PER 2018 WSEC C403.3.1.
- CONTROLS TO USE A 5°F MINIMUM DEADBAND BETWEEN HEATING AND COOLING PER 2018 WSEC C403.4.2.2.
- AUTOMATIC START AND STOP CONTROLS: PROVIDE WITH AUTOMATIC START AND STOP CONTROLS AS PER 2018 WSEC C403.4.2.3.
- HEAT PUMP (UNITARY, AIR COOLED) MICROPROCESSOR CONTROLS: PROVIDE PER 2018 WSEC C403.4.1.1.
- PROVIDE HYDRONIC BALANCING DEVICES IN ALL HEATING AND COOLING COILS AS REQUIRED BY 2018 WSEC C408.2.2.2.
- ALL HYDRONIC PIPING THAT SERVES AS PART OF A HEATING OR COOLING SYSTEM SHALL BE INSULATED IN ACCORDANCE WITH TABLE C403.10.3 OF THE 2018 WSEC. ALL GLASS FIBER PREFORMED PIPE INSULATION SHALL MEET ASTM C567.
- MOTORS THAT ARE 7.5 HP OR LARGER SHALL BE EQUIPPED WITH VARIABLE SPEED DRIVES OR OTHER MEANS OF CONTROLS TO PROVIDE FAN MOTOR DEMAND OF NO MORE THAN 30 PERCENT OF DESIGN WATTAAGE AT 50 PERCENT AIR VOLUME AND 1/3 MAXIMUM STATIC PRESSURE. 30 PERCENT OF DESIGN WATER FLOW FOR PUMPS.
- ALL HVAC EQUIPMENT SHALL MEET MINIMUM EFFICIENCY REQUIREMENTS INDICATED IN TABLES C403.3.2(1) THROUGH C403.3.2(12). WHERE MULTIPLE RATING REQUIREMENTS EXIST, THE EQUIPMENT SHALL SATISFY ALL STATED REQUIREMENTS.

GENERAL PLUMBING NOTES

- PROVIDE ALL LABOR, MATERIALS, AND EQUIPMENT NECESSARY TO CONSTRUCT A COMPLETE, OPERATIONAL PLUMBING SYSTEM FOR THE ENTIRE PROJECT AS SHOWN ON THESE DRAWINGS, INCLUDING ALL NECESSARY FEES AND PERMITS.
- THE ENTIRE INSTALLATION SHALL CONFORM TO THE MOST RECENTLY ADOPTED REQUIREMENTS OF THE PLUMBING CODE, AND ALL OTHER APPLICABLE CITY, COUNTY, STATE, AND FEDERAL CODES AND REGULATIONS IN EFFECT AT THE DATE OF THE BID. CONFORM TO ANY CODES, RULES, REGULATIONS AND REQUIREMENTS THAT THE PROJECT OWNER HAS.
- PRIOR TO FABRICATION AND INSTALLATION, THE CONTRACTOR SHALL COORDINATE THE INSTALLATION OF ALL PLUMBING WORK WITH ALL OTHER TRADES INCLUDING THE MECHANICAL CONTRACTOR, FIRE PROTECTION CONTRACTOR, ELECTRICAL CONTRACTOR, GENERAL CONTRACTOR, AND ANY CONTRACTOR HIRED DIRECTLY BY THE OWNER. WHERE CONFLICTS MAY OCCUR, THEY SHALL BE RESOLVED PRIOR TO INSTALLATION.
- THE WORKING DRAWINGS ARE DIAGRAMMATIC. BECAUSE OF THE SMALL SCALE OF THE DRAWINGS, THEY DO NOT SHOW EVERY OFFSET, BEND OR ELBOW NECESSARY FOR THE COMPLETE INSTALLATION IN THE SPACE PROVIDED. ALL LOCATIONS FOR PLUMBING EQUIPMENT AND PIPING SHALL BE CHECKED AND COORDINATED WITH THE ARCHITECTURAL, MECHANICAL, STRUCTURAL AND ELECTRICAL DRAWINGS.
- ALL EQUIPMENT SHALL BE INSTALLED IN STRICT ACCORDANCE WITH THE EQUIPMENT MANUFACTURER'S RECOMMENDATIONS. PROVIDE ALL FITTINGS, TRANSITIONS, VALVES, WORKING CLEARANCES, AND OTHER DEVICES AND ACCESSORIES REQUIRED FOR A COMPLETE, WORKABLE INSTALLATION.
- PROVIDE PROPER PROVISIONS FOR EXPANSION OR MOVEMENT OF ALL PIPING.
- PROVIDE WATER HAMMER ARRESTORS (SHOCK ABSORBERS) AT ALL PIPE LOCATIONS WHERE VALVE CLOSURES MAY CAUSE WATER HAMMER OR RESULT IN EXCESSIVE PIPE VIBRATION OR MOVEMENT. (EXAMPLES INCLUDE FLUSH VALVES, SENSOR FAUCETS, AND WASHING MACHINES).
- PROVIDE CLEANOUTS AS REQUIRED BY CODE, INCLUDING END OF RUNS.

SHEET INDEX

DWG. NO.	TITLE
M-001	MECHANICAL LEGENDS, ABBREVIATIONS AND NOTES
M-002	SCHEDULES
M-101	PARTIAL MECHANICAL BASEMENT FLOOR DEMO PLAN
M-102	PARTIAL MECHANICAL MEZZANINE DEMO PLAN
M-103	PARTIAL MECHANICAL ROOF DEMO PLAN
M-201	PARTIAL MECHANICAL BASEMENT FLOOR NEW PLAN
M-202	PARTIAL MECHANICAL MEZZANINE NEW PLAN
M-203	PARTIAL MECHANICAL ROOF NEW PLAN
M-301	SECTIONS & ELEVATIONS
M-302	SECTIONS & ELEVATIONS
M-401	SEQUENCE OF OPERATION
M-402	SEQUENCE OF OPERATION
M-501	MECHANICAL DETAILS
M-502	HEAT TRACE DETAILS



PLAZA COOLING TOWER & BOILER REPLACEMENT

701 W Riverside Ave

Spokane Transit Authority

Permit Plan Set

REV	DATE	DESCRIPTION

PROJ. NO.	200465
DRAWN	CEI
CHECKED	CEI
DATE	07/06/2021

COFFMAN ENGINEERS

SHEET TITLE:

MECHANICAL LEGENDS, ABBREVIATIONS AND NOTES

SHEET NO:

M-001

SHEET of

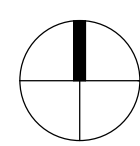
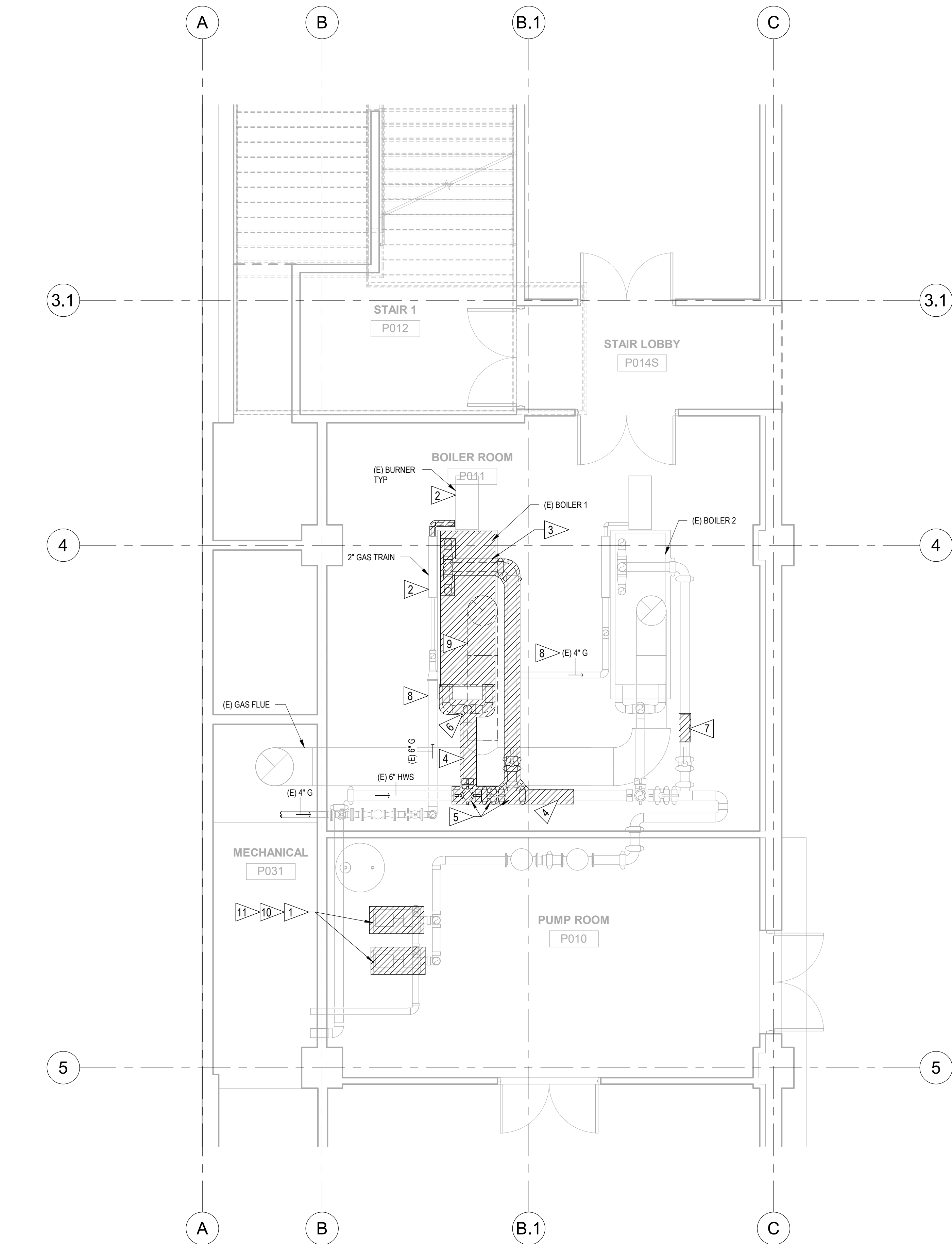
Spokane Transit
Authority

CLOSED COOLING TOWER SCHEDULE																										
TAG	LOCATION	SERVES	MANUFACTURER / MODEL	TYPE	COIL					FAN(S)			PUMP(S)			PAN HTW (KW)	MAKEUP WATER	UNIT ELECTRICAL REQUIREMENTS					DIMENSIONS (L x W x H)	OPER. WT. (LBS)	NOTES	
					FLOW RATE (GPM)	EWT (°F)	LWT (°F)	EWB (°F)	CONNECTION INLET/OUTLET	FLUID	NO.	MOTOR (HP)	CFM	NO.	MOTOR (HP)			GPM	FLA	MCA	MOCp	VOLTS				Ø
CT-1	NW ROOF	CONDENSER WATER	BAC/VFL-624-32J	CLOSED LOOP FLUID COOLER	110	90	80	68	4" / 4"	30%EG	1	7.50	17,160	1	0.50	84	3.00	1-1/2"	17.7	22	30	460	3	11ft x 4ft x 7.8ft	5,750	1,2,3,4,5,6,7
NOTES: 1. SHIPPING WEIGHT: 4,000 LBS. 2. PROVIDE WITH "BALTIQUARD PLUS" INTEGRAL VFD FAN SPEED CONTROL OR EQUIVALENT. PANEL SHIPPED LOOSE. INSTALL AS PER DEM INSTALLATION MANUAL AND COMPLIANT TO LOCAL CODE. POWER WIRING BY EC, CONTROLS WIRING BY CC. 3. PROVIDE WITH SINGLE POINT CONTROL PANEL. SCGR RATING OF 20K, 460V/3P MAIN POWER INPUT. STEPDOWN TRANSFORMER WITH INTEGRAL FUSE BLOCK, 120V ANGLARIY POWER TO CHEM FEED, LEVEL CONTROL, & MISC CONTROL REQUIREMENTS 4. PROVIDE WITH RESILIENT PAD VIBRATION ISOLATION SELECTED BY EQUIPMENT MANUFACTURER. REFER TO SPECIFICATIONS FOR ADDITIONAL INFORMATION. 5. PROVIDE NEMA-4 OUTDOOR RATED ENCLOSURE PANEL. SINGLE POINT POWER CONNECTION WITH MAIN DISCONNECT. PANEL SIZE AND WEIGHT TO BE CONFIRMED BY CONTRACTOR. 6. LEFT HAND UNIT CONFIGURATION. CONTRACTOR TO VERIFY PRIOR TO ORDERING UNIT. 7. INCLUDE START-UP AND COMMISSIONING SUPPORT AS PART OF BID.																										

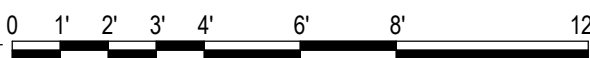
TAG	SERVICE	MFR	MODEL NUMBER	TYPE	IMP. SIZE	GPM	HEAD FT H2O	FLUID	MOTOR RPM	MOTOR SPEED	ELECTRICAL			WEIGHT (LBS)	NOTES
											VOLTS	PHASE	HP		
P-1	HEATING WATER	BELL&GOSSETT	e-1510-4GC	END SUCTION	11.125	502	120	30% P.G.	1770	CONSTANT	460	3	25	520	1, 2, 3, 4, 6
P-2	HEATING WATER	BELL&GOSSETT	e-1510-4GC	END SUCTION	11.125	502	120	30% P.G.	1770	CONSTANT	460	3	25	520	1, 2, 3, 4, 6
P-6	CONDENSER WATER	BELL&GOSSETT	e-1510-1.5AD	END SUCTION	5.5	110	120	30% E.G.	3600	CONSTANT	460	3	10	318	2, 3, 5, 6

NOTES:
1. P-1 AND P-2 ARE REDUNDANT AND SHALL ALTERNATE OPERATION.
2. PREMIUM EFFICIENCY, INVERTER DUTY RATED MOTOR. VARIABLE FREQUENCY DRIVE PROVIDED BY CONTROLS, INSTALLED BY E.C. PROVIDE SHAFT GROUNDING BY AEGIS IF NOT INCLUDED WITH PUMP.
3. PUMP SHALL HAVE SILICON CARBIDE ON VITON SEALS.
4. PROVIDE SUCTION DIFFUSER, MODEL #F: GF-3X/GF-3Z
5. PROVIDE SUCTION DIFFUSER, MODEL #F: DB-3X/DB-3Z
6. I/C TO ALIGN PUMPS PRIOR TO STARTUP.

[illegible]SHEET OF



PARTIAL MECHANICAL BASEMENT FLOOR DEMO PLAN

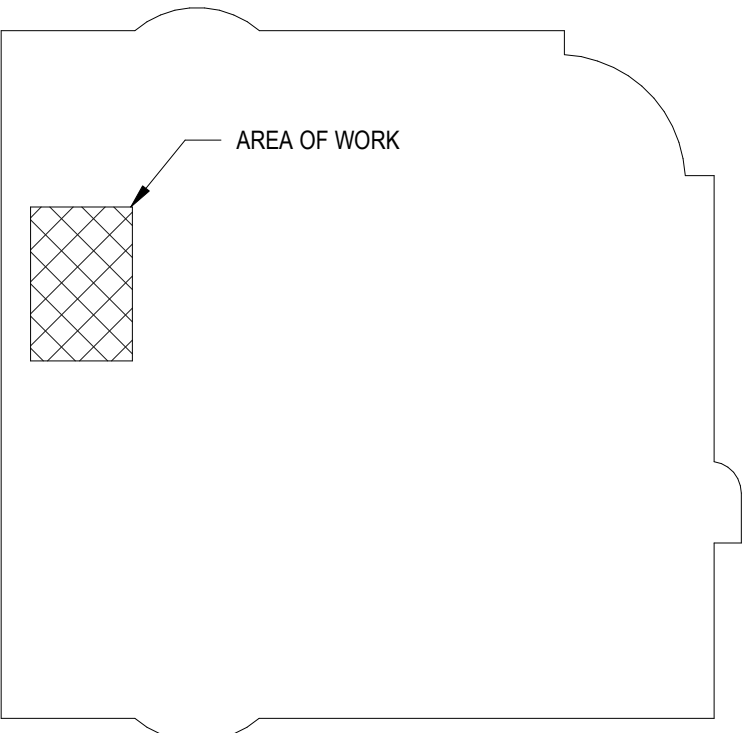


GENERAL NOTES

1. THE DRAWINGS HAVE BEEN PREPARED USING STA PROVIDED RECORD DRAWINGS AND THROUGH FIELD OBSERVATIONS. VERIFY EXISTING CONDITIONS AS THEY APPLY TO THIS PROJECT PRIOR TO BID.
2. EXISTING WORK IS SHOWN WITH LIGHT LINES AND DEMO WORK IS SHOWN HATCHED, TYP.
3. CONTRACTOR SHALL COORDINATE ANY SHUT DOWN OF SYSTEMS WITH STA.
4. CONTRACTOR SHALL BE RESPONSIBLE FOR DISPOSAL OF ALL ITEMS THE STA DOES NOT WISH TO RETAIN FOR FUTURE USE.
5. ALL EXISTING TPO ROOFING NEEDS TO BE PROTECTED FROM DAMAGE AT ALL TIMES. ANY REQUIRED REPAIRS DUE TO DAMAGE CAUSED BY THE WORK, CONTRACTOR, SUBCONTRACTORS, MATERIAL SUPPLIERS ARE THE RESPONSIBILITY OF THE CONTRACTOR.
6. CONTRACTOR IS RESPONSIBLE FOR ALL FALL PROTECTION MEANS AND METHODS REQUIRED TO PERFORM WORK.
7. CONTRACTOR TO COORDINATE STORAGE AND STAGING OF ALL EQUIPMENT AND MATERIALS WITH STA.
8. BOILER PIPING, NEW VALVES AND DRAFT DAMPER ARE TO BE INSTALLED IMMEDIATELY AFTER PIPING DEMOLITION IN ORDER TO MINIMIZE THE SHUTDOWN OF BOILER 2. BOILER 2 IS TO BE AVAILABLE FOR USE EXCEPT FOR BRIEF PERIODS WHEN HEATING IS NOT REQUIRED AND AS SCHEDULED WITH STA. SEE DRAWING M-201 FOR NEW VALVE AND DRAFT DAMPER LOCATIONS.

SHEET NOTES

- 1 DEMO EXISTING HEATING HOT WATER PUMPS BACK TO FLANGES. ALL PIPING AND VALVES TO REMAIN. PREP FOR RECONNECTION TO NEW PUMPS. DEMO EXISTING SUCTION DIFFUSER.
- 2 BURNER AND GAS TRAIN TO BE REMOVED AND SAVED FOR REINSTALLATION ON NEW BOILER.
- 3 REMOVE (E) BOILER. HOUSEKEEPING PAD TO REMAIN.
- 4 REMOVE 6\"/>
- 5 REMOVE TEES AND BUTTERFLY VALVES. SEE GENERAL NOTE 8.
- 6 REMOVE HWS PIPING AND 8\"/>
- 7 REMOVE PIPE FOR INSTALLATION OF NEW TEE AND VALVE. PIPING CAN BE REMOVED BACK TO NEAREST COUPLING. SEE GENERAL NOTE 8.
- 8 GAS PIPING TO REMAIN.
- 9 REMOVE BOILER BREECHING TO TEE FITTING. SEE GENERAL NOTE 8.
- 10 EXISTING CONCRETE PAD TO BE RE-USED OR EXTENDED AS REQUIRED TO SERVICE NEW PUMP.
- 11 DEMO EXISTING CONTROL WIRING AND CONDUIT ASSOCIATED WITH PUMP OR RE-USE IF IN GOOD CONDITION.



BASEMENT KEYPLAN



PLAZA COOLING
TOWER & BOILER
REPLACEMENT

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Authority

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SHEET TITLE:

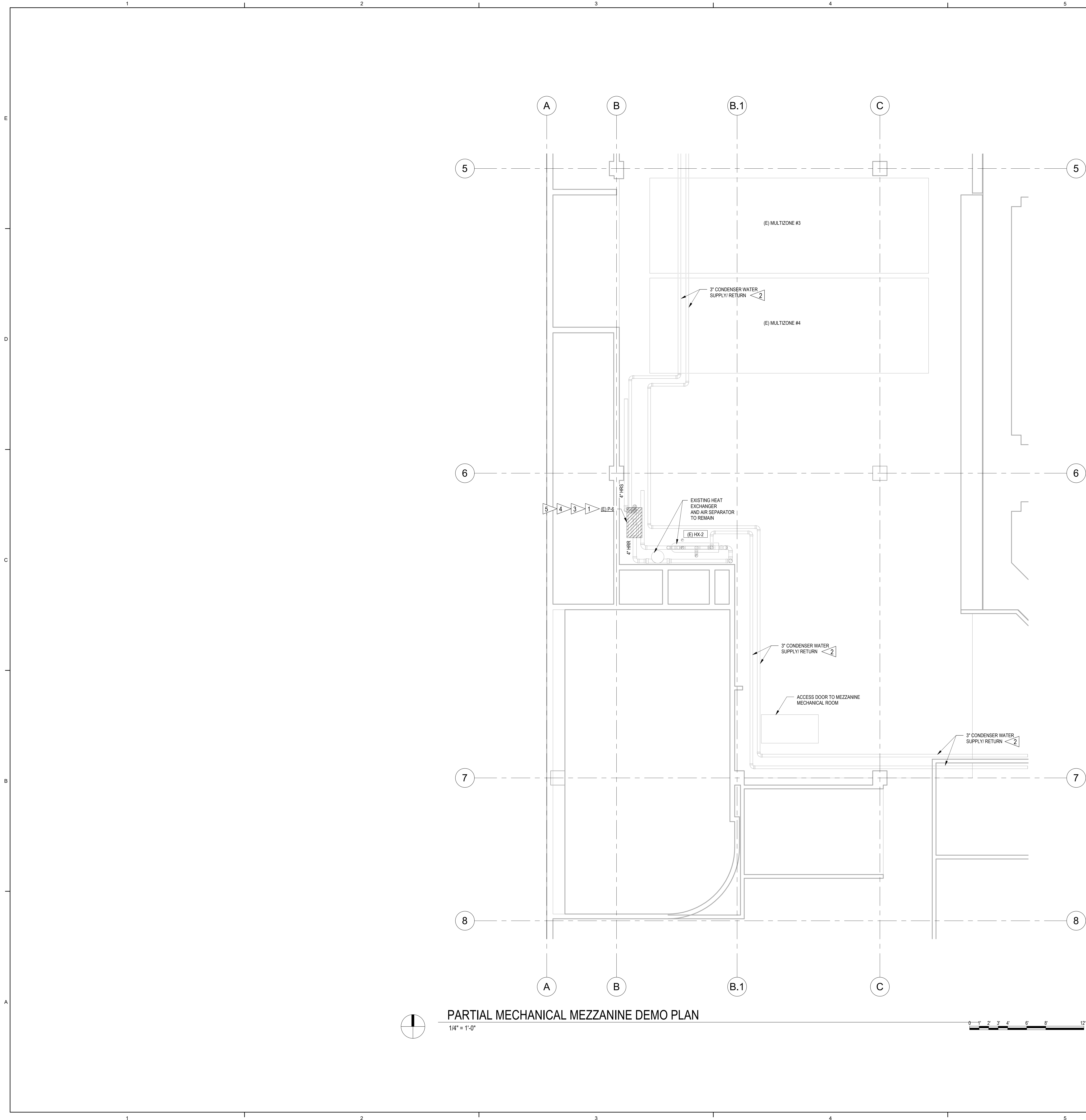
PARTIAL
MECHANICAL
BASEMENT FLOOR
DEMO PLAN

SHEET NO:

M-101

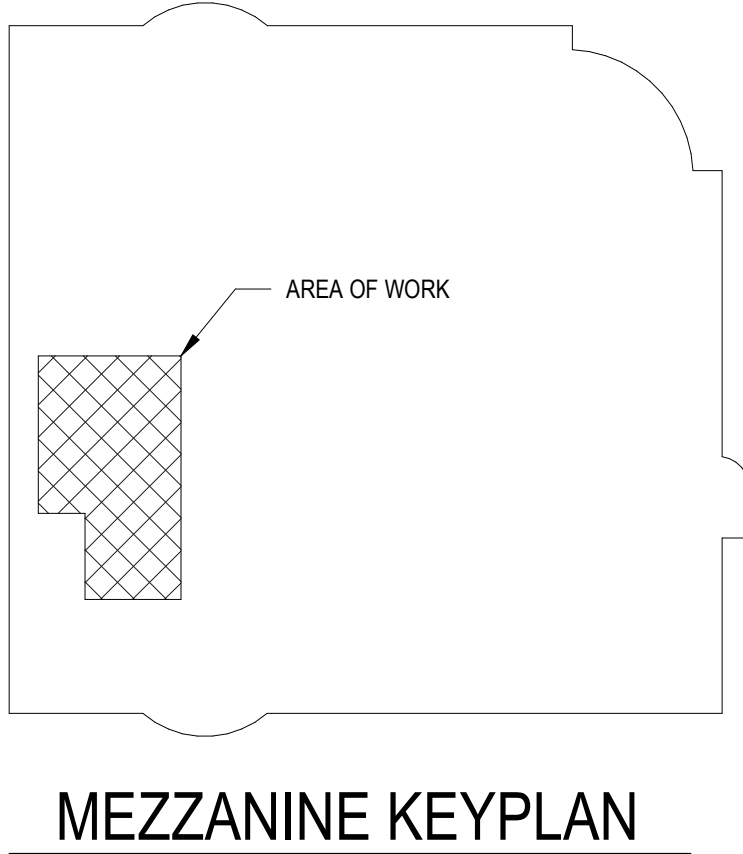
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- GENERAL NOTES**
1. THE DRAWINGS HAVE BEEN PREPARED USING STA PROVIDED RECORD DRAWINGS AND THROUGH FIELD OBSERVATIONS. VERIFY EXISTING CONDITIONS AS THEY APPLY TO THIS PROJECT PRIOR TO BID.
 2. EXISTING WORK IS SHOWN WITH LIGHT LINES AND DEMO WORK IS SHOWN HATCHED, TYP.
 3. CONTRACTOR SHALL COORDINATE ANY SHUT DOWN OF SYSTEMS WITH STA.
 4. CONTRACTOR SHALL BE RESPONSIBLE FOR DISPOSAL OF ALL ITEMS THE STA DOES NOT WISH TO RETAIN FOR FUTURE USE.
 5. ALL EXISTING TPO ROOFING NEEDS TO BE PROTECTED FROM DAMAGE AT ALL TIMES. ANY REQUIRED REPAIRS DUE TO DAMAGE CAUSED BY THE WORK, CONTRACTOR, SUBCONTRACTORS, MATERIAL SUPPLIERS ARE THE RESPONSIBILITY OF THE CONTRACTOR.
 6. CONTRACTOR IS RESPONSIBLE FOR ALL FALL PROTECTION MEANS AND METHODS REQUIRED TO PERFORM WORK.
 7. CONTRACTOR TO COORDINATE STORAGE AND STAGING OF ALL EQUIPMENT AND MATERIALS WITH STA.

- SHEET NOTES**
- 1 DEMO EXISTING PUMP P-6. RE-USE PIPING AND VALVES AND PREP FOR CONNECTION TO NEW PUMP. DEMO EXISTING SUCTION DIFFUSER.
 - 2 EXISTING PIPING TO REMAIN FOR RE-USE.
 - 3 CONTRACTOR TO FIELD VERIFY ADEQUATE SPACE REQUIREMENTS FOR REMOVAL OF OLD PUMP.
 - 4 EXISTING CONCRETE PAD TO BE RE-USED OR EXTENDED AS REQUIRED TO SERVICE NEW PUMP.
 - 5 DEMO EXISTING CONTROL WIRING AND CONDUIT ASSOCIATED WITH PUMP OR -RE-USE IF GOOD CONDITION.



**PLAZA COOLING
TOWER & BOILER
REPLACEMENT**

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REV	DATE	DESCRIPTION

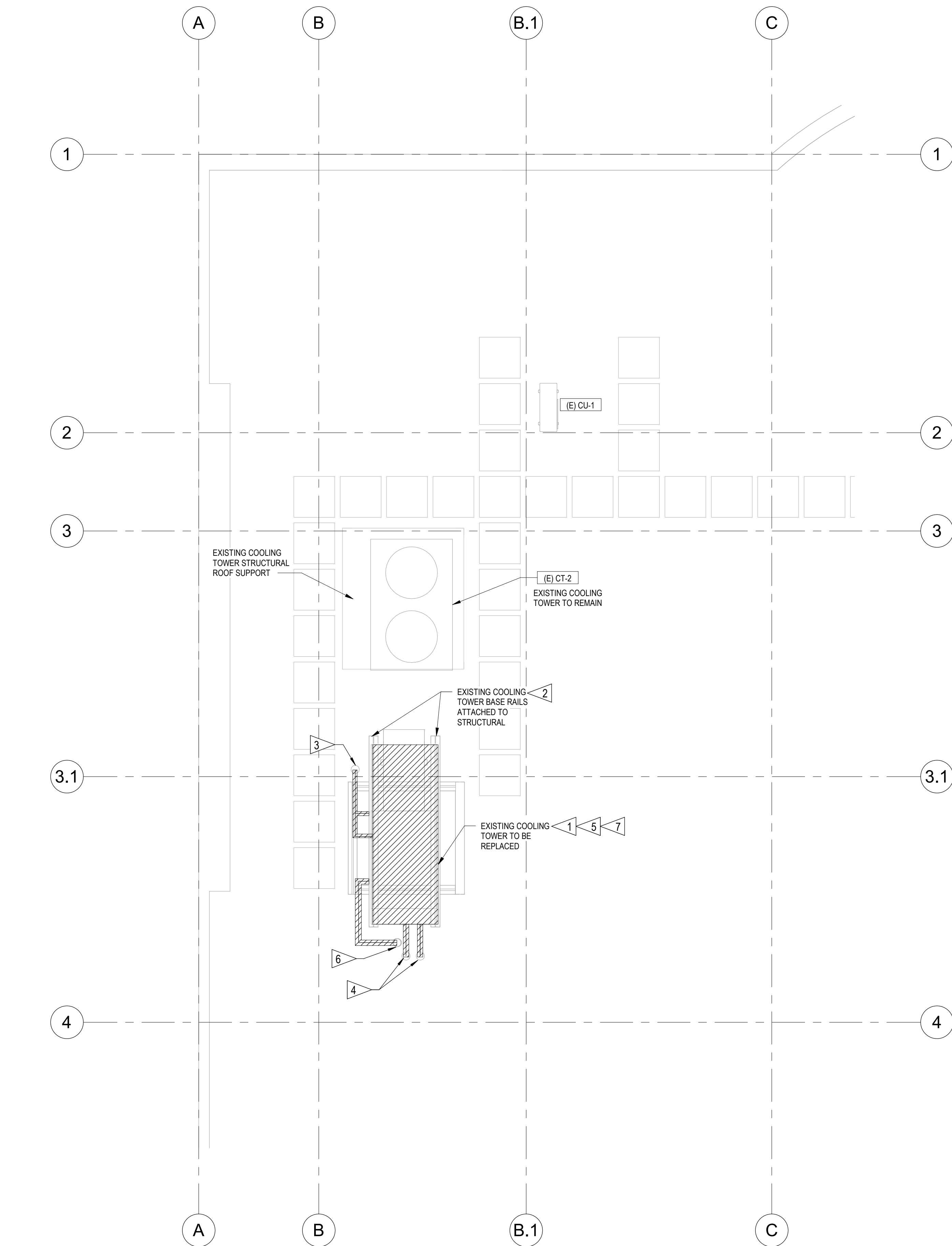
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DATE	07/06/2021

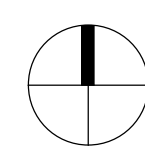
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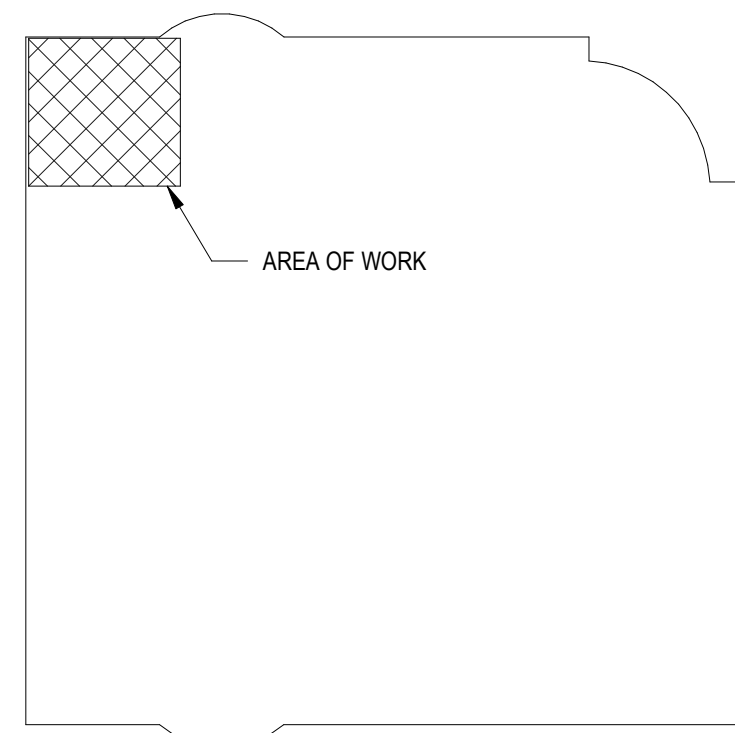
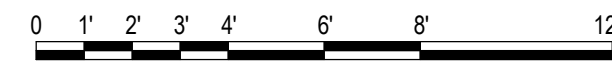
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PARTIAL
MECHANICAL
MEZZANINE DEMO
PLAN

SHEET NO:
M-102
SHEET of

7/6/2021 3:05:52 PM



 PARTIAL MECHANICAL ROOF DEMO PLAN
1/4" = 1'-0"



ROOF KEYPLAN

GENERAL NOTES

1. THE DRAWINGS HAVE BEEN PREPARED USING STA PROVIDED RECORD DRAWINGS AND THROUGH FIELD OBSERVATIONS. VERIFY EXISTING CONDITIONS AS THEY APPLY TO THIS PROJECT PRIOR TO BID.
2. EXISTING WORK IS SHOWN WITH LIGHT LINES AND DEMO WORK IS SHOWN HATCHED, TYP.
3. CONTRACTOR SHALL COORDINATE ANY SHUT DOWN OF SYSTEMS WITH STA.
4. CONTRACTOR SHALL BE RESPONSIBLE FOR DISPOSAL OF ALL ITEMS THE STA DOES NOT WISH TO RETAIN FOR FUTURE USE.
5. ALL EXISTING TPO ROOFING NEEDS TO BE PROTECTED FROM DAMAGE AT ALL TIMES. ANY REQUIRED REPAIRS DUE TO DAMAGE CAUSED BY THE WORK, CONTRACTOR, SUBCONTRACTORS, MATERIAL SUPPLIERS ARE THE RESPONSIBILITY OF THE CONTRACTOR.
6. CONTRACTOR IS RESPONSIBLE FOR ALL FALL PROTECTION MEANS AND METHODS REQUIRED TO PERFORM WORK.
7. CONTRACTOR TO COORDINATE STORAGE AND STAGING OF ALL EQUIPMENT AND MATERIALS WITH STA.

COOLING TOWER STRUCTURAL BASE AND EXISTING EXPOSED PIPING REFURBISHMENT NOTE

1. PREPARE ALL DAMAGED/MISSING PAINT PER SSPC-SP 15, COMMERCIAL GRADE POWER TOOL CLEANING. THE PREPARED AREA SHALL EXTEND AT LEAST 2 INCHES INTO ADJACENT TIGHTLY ADHERING, INTACT COATING. PROVIDE ONE PRIMER COAT OF MILL WHITE COLOR, SHERWIN WILLIAMS MACROPOXY 646 FAST CURE EPOXY OR EQUIVALENT. PROVIDE TWO COATS OF SHERWIN WILLIAMS SHERLOXANE 500 HIGH PERFORMANCE EPOXY SILOXANE HYBRID OR EQUIVALENT. EACH COAT SHALL HAVE A MINIMUM THICKNESS OF 4.7 MILS AND SHALL BE FULLY CURED PRIOR TO APPLICATION OF SECOND COAT. MATCH COLOR OF EXISTING STRUCTURAL BASE PAINT.

SHEET NOTES

- 1 DEMO EXISTING COOLING TOWER.
- 2 RE-USE AND REFURBISH EXISTING COOLING TOWER MOUNTING RAILS AND STRUCTURAL BASE. SEE COOLING TOWER STRUCTURAL BASE AND EXISTING EXPOSED PIPING REFURBISHMENT NOTE.
- 3 DEMO EXISTING COOLING TOWER DRAIN AND BLOW DOWN PIPING BACK TO A FOOT ABOVE EXISTING ROOF LEVEL PENETRATION. PREP DRAIN PIPE TO RECONNECT NEW COOLING TOWER DRAIN.
- 4 DISCONNECT EXISTING 3" DIAMETER CONDENSER WATER LINES AT EXISTING COOLING TOWER AND REMOVE PIPES BACK TO A FOOT ABOVE ROOF SERVICE.
- 5 INSTALLING CONTRACTOR TO COORDINATE DEMO OF EXISTING COOLING TOWER WITH STA. ANY CRANE LAY-DOWN SHALL BE A MINIMUM 20' FROM BUILDING EDGE AND ANY SIDEWALK, TRAFFIC DISRUPTIONS SHALL BE OUTSIDE OF REGULAR STA PLAZA HOURS OF OPERATION AND MOST LIKELY AT NIGHT.
- 6 DISCONNECT EXISTING MAKEUP WATER LINE AT COOLING TOWER, AND PREP CONNECTION FOR REUSE. REUSE PIPE, HOSE SPIGOT, WATER HAMMER ARRESTOR, AND RPBP.
- 7 DEMO EXISTING CONTROLS WIRING AND ELECTRICAL CONDUIT ASSOCIATED WITH EXISTING COOLING TOWER.



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SHEET TITLE:

**PARTIAL
MECHANICAL ROOF
DEMO PLAN**

SHEET NO:

M-103

SHEET of



PLAZA COOLING
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COFFMAN ENGINEERS

SHEET TITLE:

PARTIAL
MECHANICAL
BASEMENT FLOOR
NEW PLAN

SHEET NO:

M-201

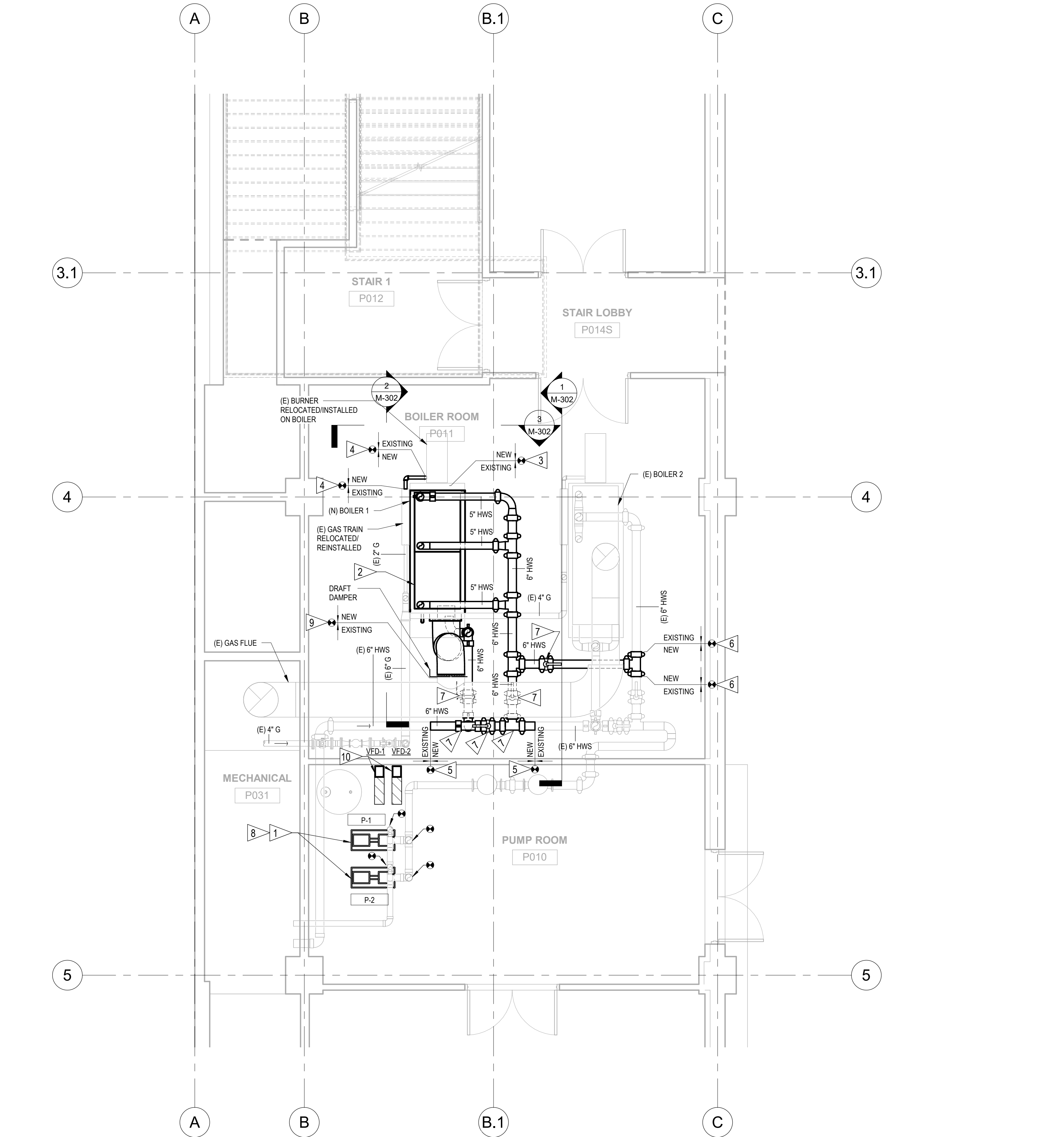
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GENERAL NOTES

- THE DRAWINGS HAVE BEEN PREPARED USING STA PROVIDED RECORD DRAWINGS AND THROUGH FIELD OBSERVATIONS. VERIFY EXISTING CONDITIONS AS THEY APPLY TO THIS PROJECT PRIOR TO BID.
- EXISTING WORK IS SHOWN WITH LIGHT LINES AND DEMO WORK IS SHOWN HATCHED, TYP.
- CONTRACTOR SHALL COORDINATE ANY SHUT DOWN OF SYSTEMS WITH STA.
- CONTRACTOR SHALL BE RESPONSIBLE FOR DISPOSAL OF ALL ITEMS THE STA DOES NOT WISH TO RETAIN FOR FUTURE USE.
- ALL EXISTING TPO ROOFING NEEDS TO BE PROTECTED FROM DAMAGE AT ALL TIMES. ANY REQUIRED REPAIRS DUE TO DAMAGE CAUSED BY THE WORK. CONTRACTOR, SUBCONTRACTORS, MATERIAL SUPPLIERS ARE THE RESPONSIBILITY OF THE CONTRACTOR.
- CONTRACTOR IS RESPONSIBLE FOR ALL FALL PROTECTION MEANS AND METHODS REQUIRED TO PERFORM WORK.
- CONTRACTOR TO COORDINATE STORAGE AND STAGING OF ALL EQUIPMENT AND MATERIALS WITH STA.
- BOILER PIPING, NEW VALVES AND DRAFT DAMPER ARE TO BE INSTALLED IMMEDIATELY AFTER PIPING DEMOLITION IN ORDER TO MINIMIZE THE SHUTDOWN OF BOILER 2. BOILER 2 IS TO BE AVAILABLE FOR USE EXCEPT FOR BRIEF PERIODS WHEN HEATING IS NOT REQUIRED AND AS SCHEDULED WITH STA. SEE DRAWING M-201 FOR NEW VALVE AND DRAFT DAMPER LOCATIONS.

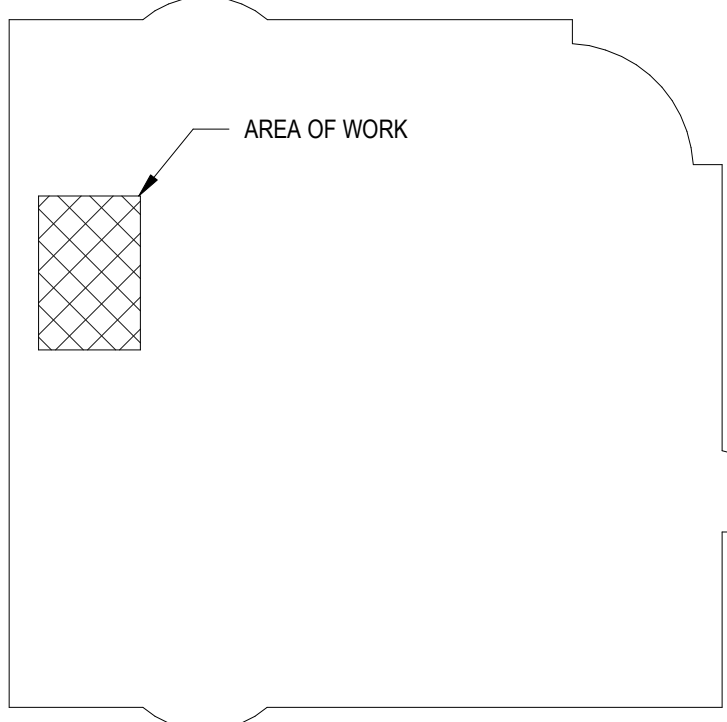
SHEET NOTES

- PROVIDE NEW HEATING WATER PUMP. INSTALL ON EXISTING HOUSEKEEPING PAD AND CONNECT TO EXISTING PIPING AND VALVES. FURNISH WITH VFD AND END-SUCTION DIFFUSER. MOUNT PER MANUFACTURER'S INSTRUCTIONS.
- INSTALL BOILER ON EXISTING HOUSEKEEPING PAD.
- INSTALL EXISTING BURNER ON NEW BOILER.
- RE-INSTALL GAS TRAIN AND CONNECT TO BURNER.
- CONNECT NEW 6" PIPING TO EXISTING. TIE IN CAN BE MADE AT NEAREST COUPLING. SEE GENERAL NOTE 8.
- INSTALL NEW 6" TEE IN EXISTING PIPE. SEE GENERAL NOTE 8.
- INSTALL NEW TEES AND BUTTERFLY VALVES. SEE GENERAL NOTE 8.
- CONTRACTOR SHALL REUSE EXISTING CONCRETE PAD. EXTEND AND PATCH PAD AS NEEDED TO ALLOW FOR PROPER INSTALLATION OF NEW PUMPS.
- ROUTE NEW BOILER BREECHING TO EXISTING TEE FITTING. PROVIDE NEW DRAFT DAMPER AT TIE IN POINT. SEE NOTE 8.
- PROVIDE NEW VFD FOR HEATING WATER PUMP. MOUNT TO WALL PER MANUFACTURERS RECOMMENDATIONS AND PROVIDE MINIMUM 42" OF CLEARANCE ALONG FRONT OF UNIT. CONTRACTOR TO VERIFY MOUNTING HEIGHT BASED ON AVAILABLE SPACE ON THE WALL.

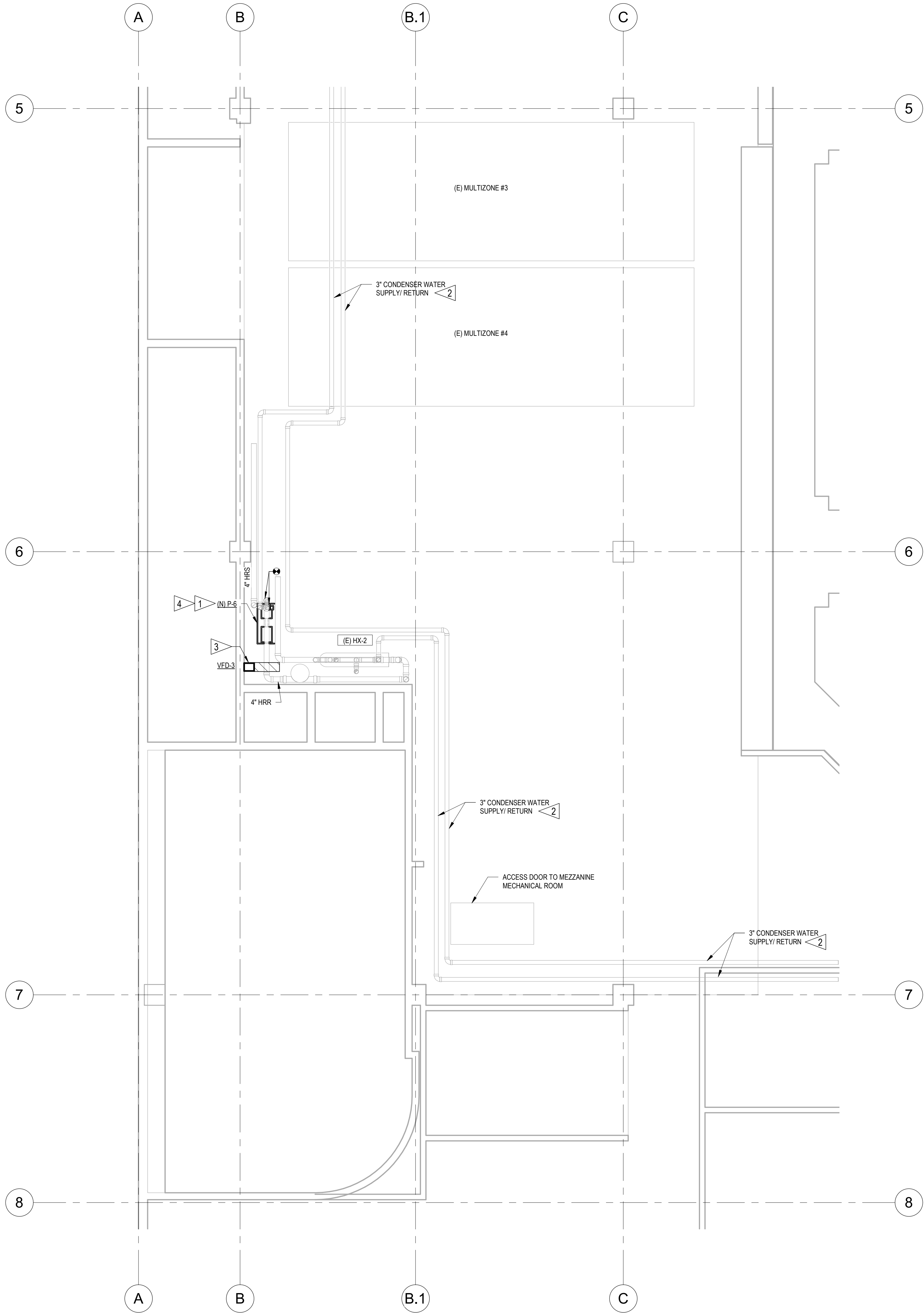


PARTIAL MECHANICAL BASEMENT FLOOR NEW PLAN
1/4" = 1'-0"

0 1' 2' 3' 4' 6' 8' 12'

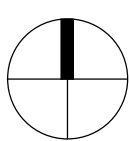


BASEMENT KEYPLAN



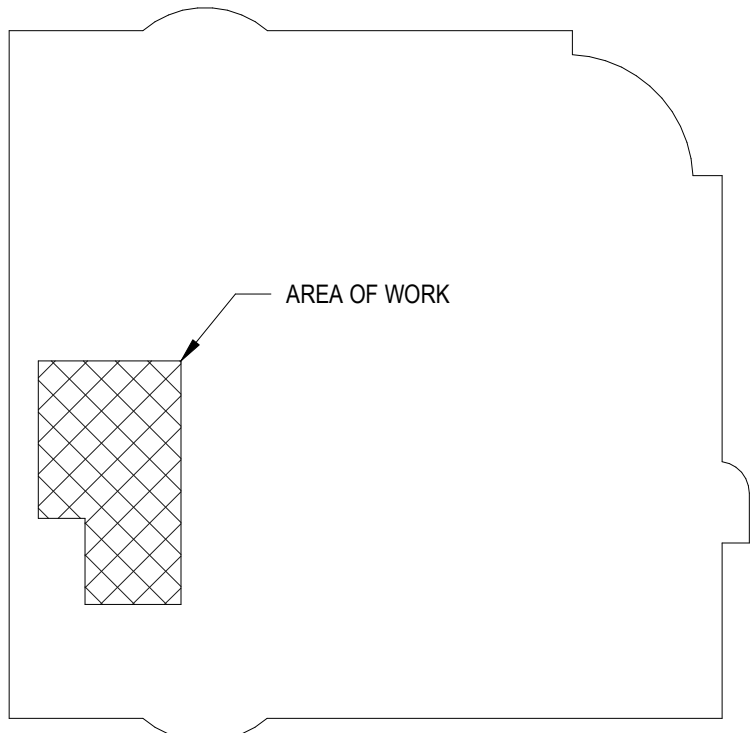
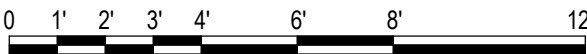
- GENERAL NOTES**
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 2. EXISTING WORK IS SHOWN WITH LIGHT LINES AND DEMO WORK IS SHOWN HATCHED, TYP.
 3. CONTRACTOR SHALL COORDINATE ANY SHUT DOWN OF SYSTEMS WITH STA.
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 5. ALL EXISTING TPO ROOFING NEEDS TO BE PROTECTED FROM DAMAGE AT ALL TIMES. ANY REQUIRED REPAIRS DUE TO DAMAGE CAUSED BY THE WORK, CONTRACTOR, SUBCONTRACTORS, MATERIAL SUPPLIERS ARE THE RESPONSIBILITY OF THE CONTRACTOR.
 6. CONTRACTOR IS RESPONSIBLE FOR ALL FALL PROTECTION MEANS AND METHODS REQUIRED TO PERFORM WORK.
 7. CONTRACTOR TO COORDINATE STORAGE AND STAGING OF ALL EQUIPMENT AND MATERIALS WITH STA.

- SHEET NOTES**
1. INSTALL NEW PUMP P-6 AND END SUCTION DIFFUSER. RE-CONNECT EXISTING PIPING TO NEW PUMP AND RE-USE EXISTING VALVES.
 2. EXISTING PIPING TO REMAIN FOR RE-USE.
 3. PROVIDE NEW VFD FOR CONDENSER WATER PUMP. MOUNT TO WALL PER MANUFACTURERS RECOMMENDATIONS AND PROVIDE MINIMUM 42" OF CLEARANCE ALONG FRONT OF UNIT. CONTRACTOR TO VERIFY MOUNTING HEIGHT BASED ON AVAILABLE SPACE ON THE WALL.
 4. CONTRACTOR SHALL REUSE EXISTING CONCRETE PAD. EXTEND AND PATCH PAD AS NEEDED TO ALLOW FOR PROPER INSTALLATION OF NEW PUMPS.



PARTIAL MECHANICAL MEZZANINE NEW PLAN

1/4" = 1'-0"



MEZZANINE KEYPLAN



PLAZA COOLING
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PARTIAL
MECHANICAL
MEZZANINE NEW
PLAN

SHEET NO:

M-202

SHEET of



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GENERAL NOTES

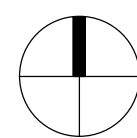
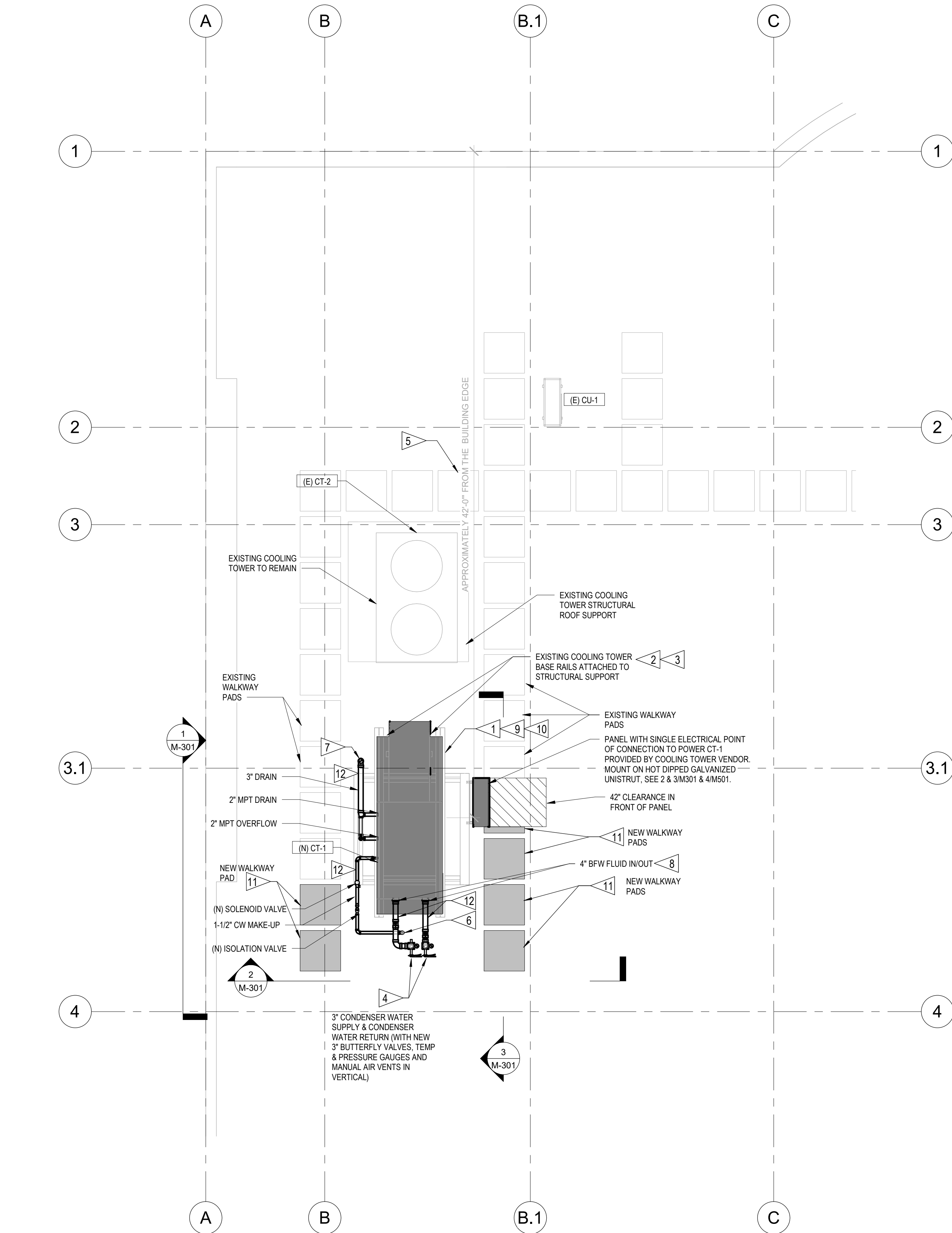
1. THE DRAWINGS HAVE BEEN PREPARED USING STA PROVIDED RECORD DRAWINGS AND THROUGH FIELD OBSERVATIONS. VERIFY EXISTING CONDITIONS AS THEY APPLY TO THIS PROJECT PRIOR TO BID.
2. EXISTING WORK IS SHOWN WITH LIGHT LINES AND DEMO WORK IS SHOWN HATCHED, TYP.
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6. CONTRACTOR IS RESPONSIBLE FOR ALL FALL PROTECTION MEANS AND METHODS REQUIRED TO PERFORM WORK.
7. CONTRACTOR TO COORDINATE STORAGE AND STAGING OF ALL EQUIPMENT AND MATERIALS WITH STA.

COOLING TOWER STRUCTURAL BASE AND EXISTING EXPOSED
PIPING REFURBISHMENT NOTE

1. PREPARE ALL DAMAGED/MISSING PAINT PER SSPC-SP 15, COMMERCIAL GRADE POWER TOOL CLEANING. THE PREPARED AREA SHALL EXTEND AT LEAST 2 INCHES INTO ADJACENT TIGHTLY ADHERING, INTACT COATING. PROVIDE ONE PRIMER COAT OF MILL WHITE COLOR, SHERWIN WILLIAMS MACROPOXY 646 FAST CURE EPOXY OR EQUIVALENT. PROVIDE TWO COATS OF SHERWIN WILLIAMS SHER-LOXANE 800 HIGH-PERFORMANCE EPOXY SLOVANE HYBRID OR EQUIVALENT. EACH COAT SHALL HAVE A MINIMUM THICKNESS OF 4.7 MILS AND SHALL BE FULLY CURED PRIOR TO APPLICATION OF SECOND COAT. MATCH COLOR OF EXISTING STRUCTURAL BASE PAINT.

SHEET NOTES

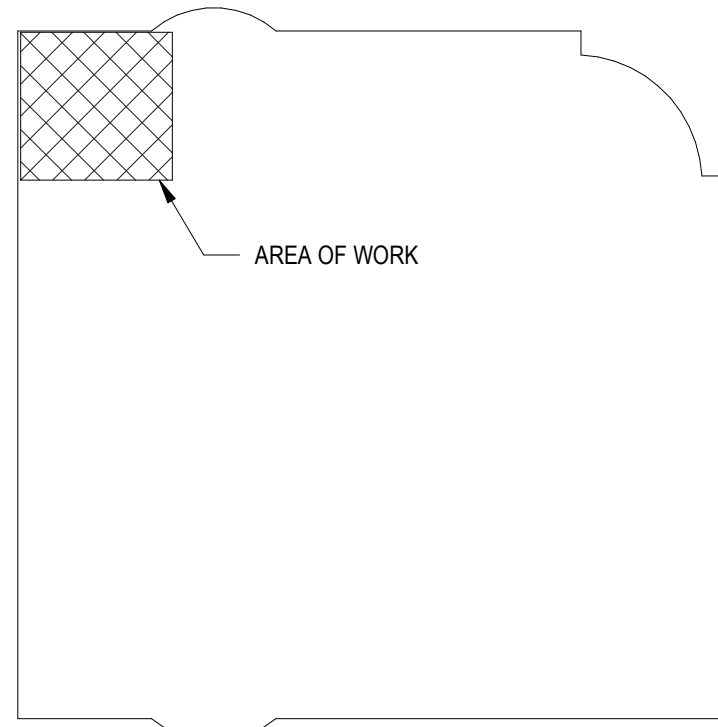
1. INSTALL NEW COOLING TOWER AND VIBRATION ISOLATORS ON EXISTING RAILS. REUSE BOLT HOLES TO THE MAXIMUM EXTENT POSSIBLE.
2. REUSE AND REFURBISH EXISTING COOLING TOWER STRUCTURAL BASE. SEE COOLING TOWER STRUCTURAL BASE AND EXISTING EXPOSED PIPING REFURBISHMENT NOTE.
3. REUSE AND REFURBISH EXISTING COOLING TOWER MOUNTING RAILS. SEE COOLING TOWER STRUCTURAL BASE AND EXISTING EXPOSED PIPING REFURBISHMENT NOTE.
4. EXTEND EXISTING 3" DIAMETER CONDENSER WATER LINES TO NEW COOLING TOWER CONNECTION POINT. INSTALL 4" TRANSITIONS TO CONNECT TO NEW COOLING TOWER INLET/OUTLET. REUSE EXISTING CONDENSER WATER LINES, INSULATION AND ROOF PENETRATIONS TO THE MAXIMUM EXTENT POSSIBLE. PATCH ROOF AND INSTALL NEW PIPE INSULATION, AS NECESSARY.
5. INSTALLING CONTRACTOR TO COORDINATE INSTALLATION OF NEW WITH STA. ANY CRANE LAY-DOWN SHALL BE A MINIMUM 20' FROM BUILDING EDGE AND ANY SIDEWALK, TRAFFIC AND SIDEWALK DISRUPTIONS SHALL BE OUTSIDE OF REGULAR STA PLAZA HOURS OF OPERATION AND MOST LIKELY AT NIGHT.
6. EXTEND EXISTING MAKEUP WATER LINE TO NEW COOLING TOWER. REUSE EXISTING PIPE, HOSE SPISOT AND WATER HAMMER ARRESTOR IN PLACE. INSTALL NEW FILL WATER SOLENOID VALVE, PATCH ROOF AND INSTALL NEW PIPE INSULATION AS NECESSARY.
7. REUSE EXISTING DRAIN PIPE AND EXTEND TO NEW COOLING TOWER DRAIN AND BLOW DOWN CONNECTION POINTS.
8. HEAT TRACE CW MAKE-UP LINE. SEE SHEET M502 FOR HEAT TRACE DETAILS.
9. CONTRACTOR TO PROVIDE SHOP DRAWINGS OF COOLING TOWER INSTALLATION TO ENGINEER OF RECORD FOR REVIEW AND APPROVAL.
10. CONTRACTOR TO COMPARE NEW COOLING TOWER PIPING CONNECTION LOCATIONS TO EXISTING PIPE ROUTING AND VERIFY ANY MODIFICATIONS THAT NEED TO BE MADE TO EXISTING PIPING IN ORDER TO CONNECT TO NEW COOLING TOWER. IF COOLING TOWER IS DIFFERENT FROM SCHEDULED BASIS OF DESIGN, CONTRACTOR TO CONFIRM IF MODIFICATIONS TO EXISTING PIPING ARE NECESSARY AND SUBMIT TO ENGINEER OF RECORD FOR REVIEW AND APPROVAL.
11. CONTRACTOR TO VERIFY IF ADDITIONAL WALKWAY PADS ARE REQUIRED AND PROVIDE AS NECESSARY TO EXTEND ALONG LENGTH OF EQUIPMENT.
12. SUPPORT ALL PIPING FROM ROOF AS NEEDED.



PARTIAL MECHANICAL ROOF NEW PLAN

1/4" = 1'-0"

0 1' 2' 3' 4' 6' 8' 12'



ROOF KEYPLAN

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SHEET TITLE:

PARTIAL
MECHANICAL ROOF
NEW PLAN

SHEET NO:

M-203

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COFFMAN ENGINEERS

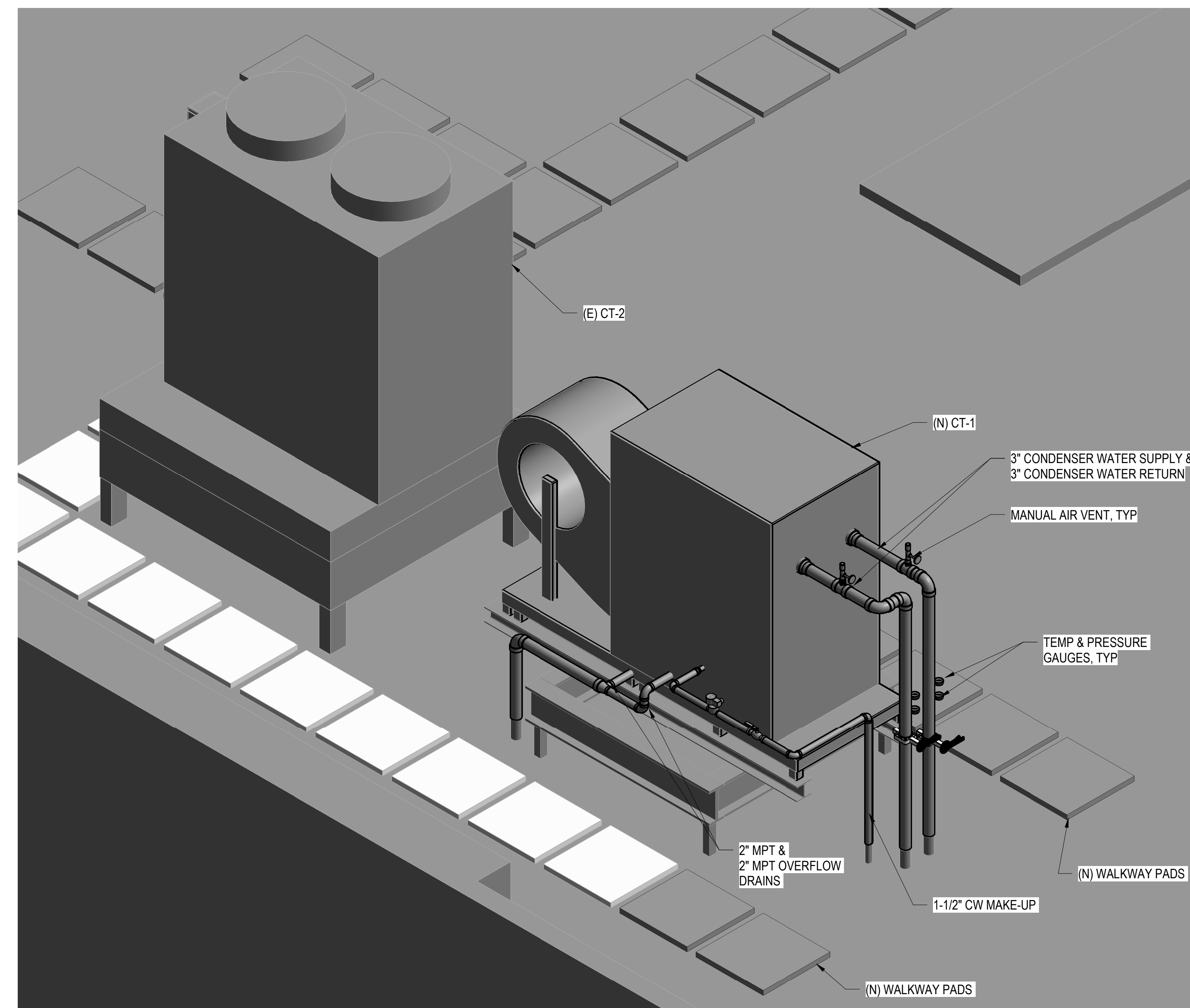
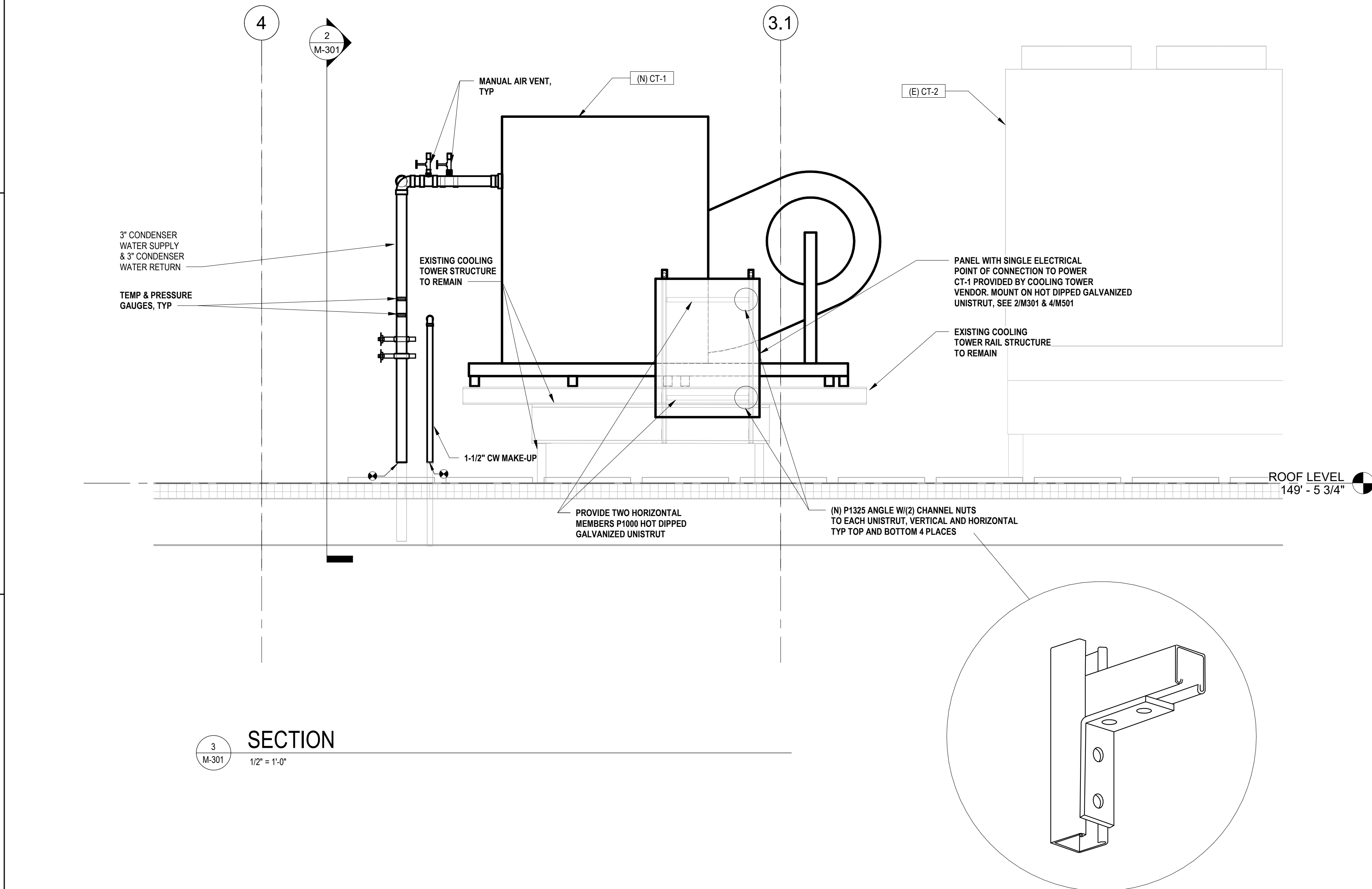
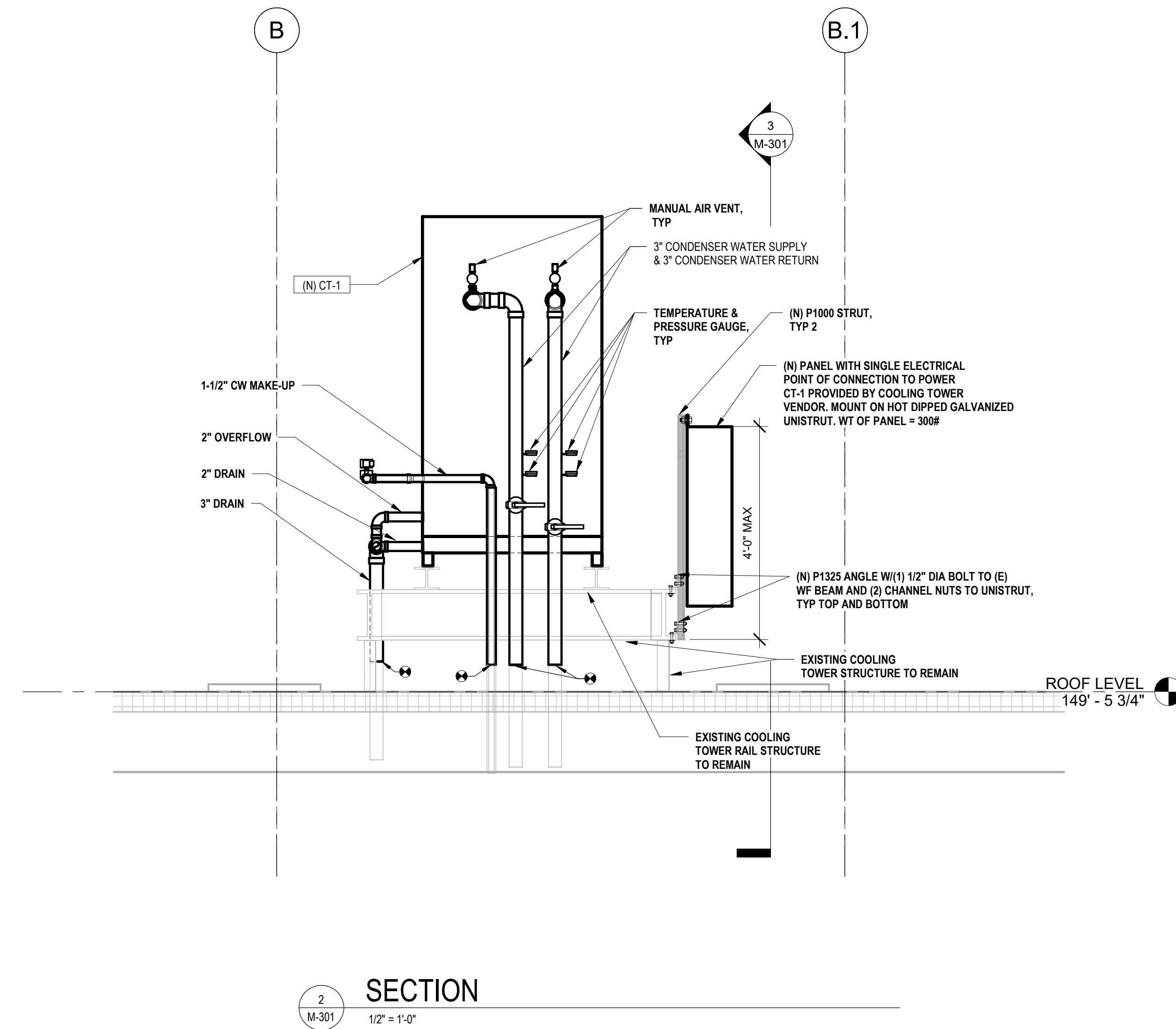
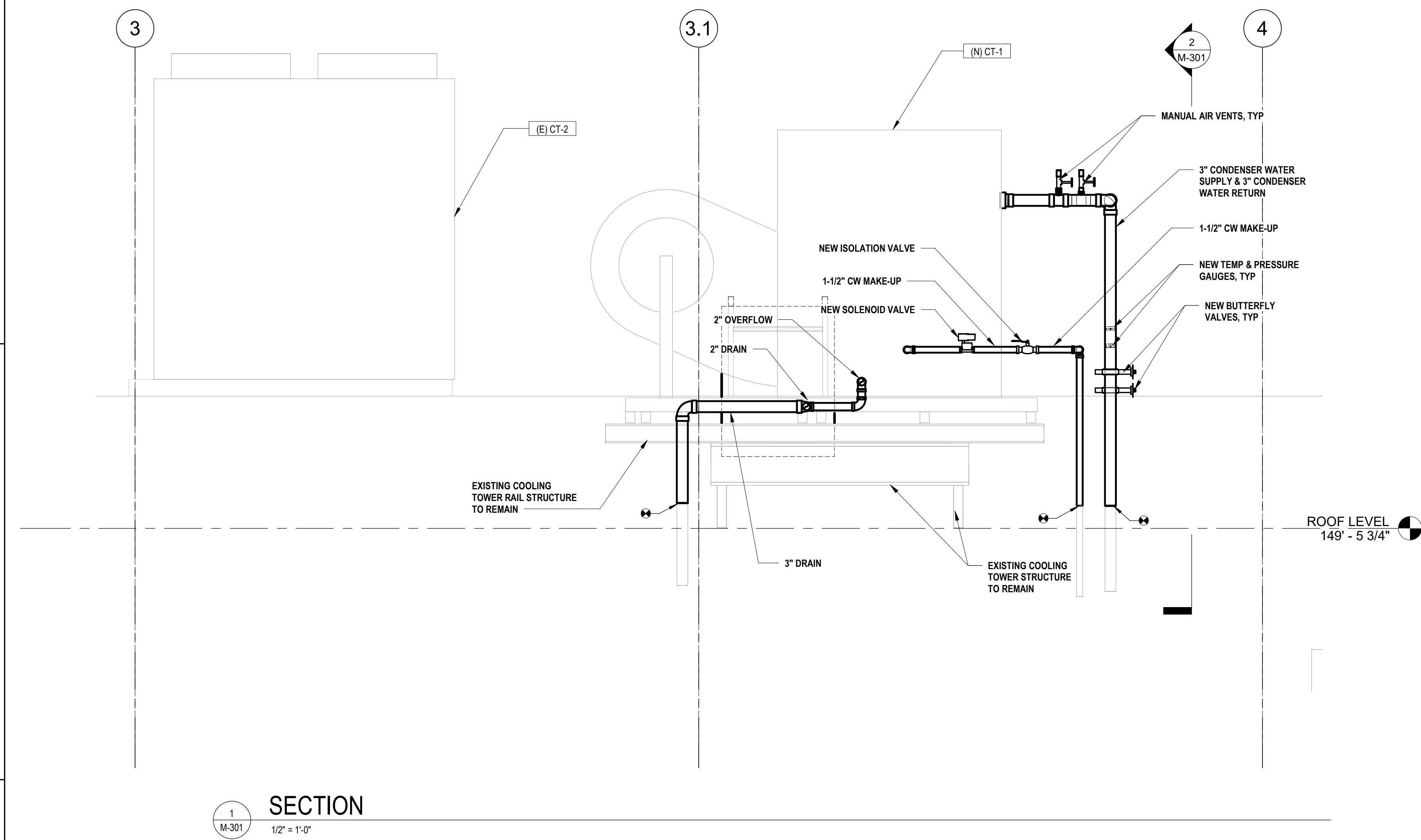
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SECTIONS &
ELEVATIONS

SHEET NO:

M-301

SHEET of





PLAZA COOLING
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COFFMAN ENGINEERS

SHEET TITLE:

SECTIONS &
ELEVATIONS

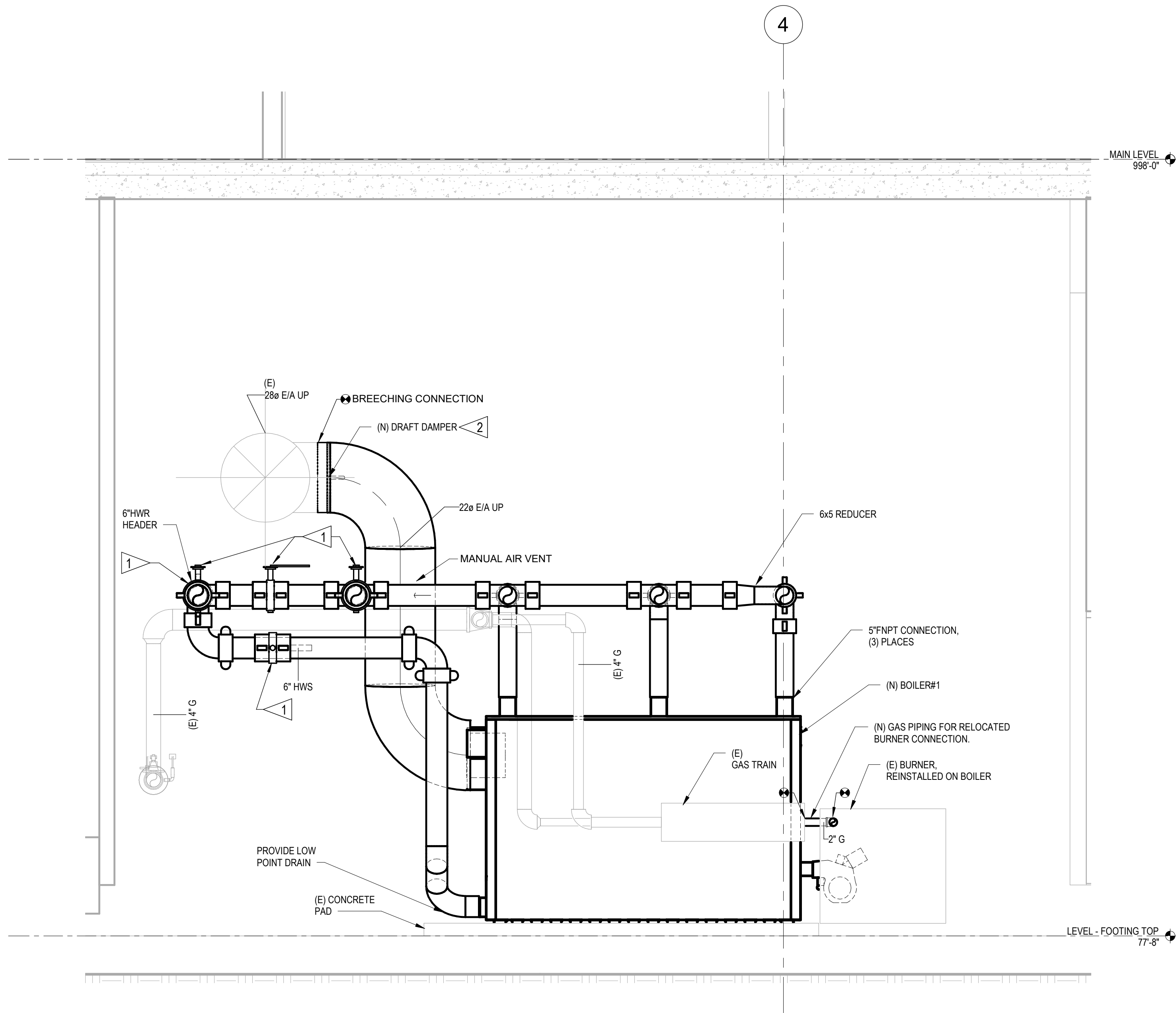
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M-302

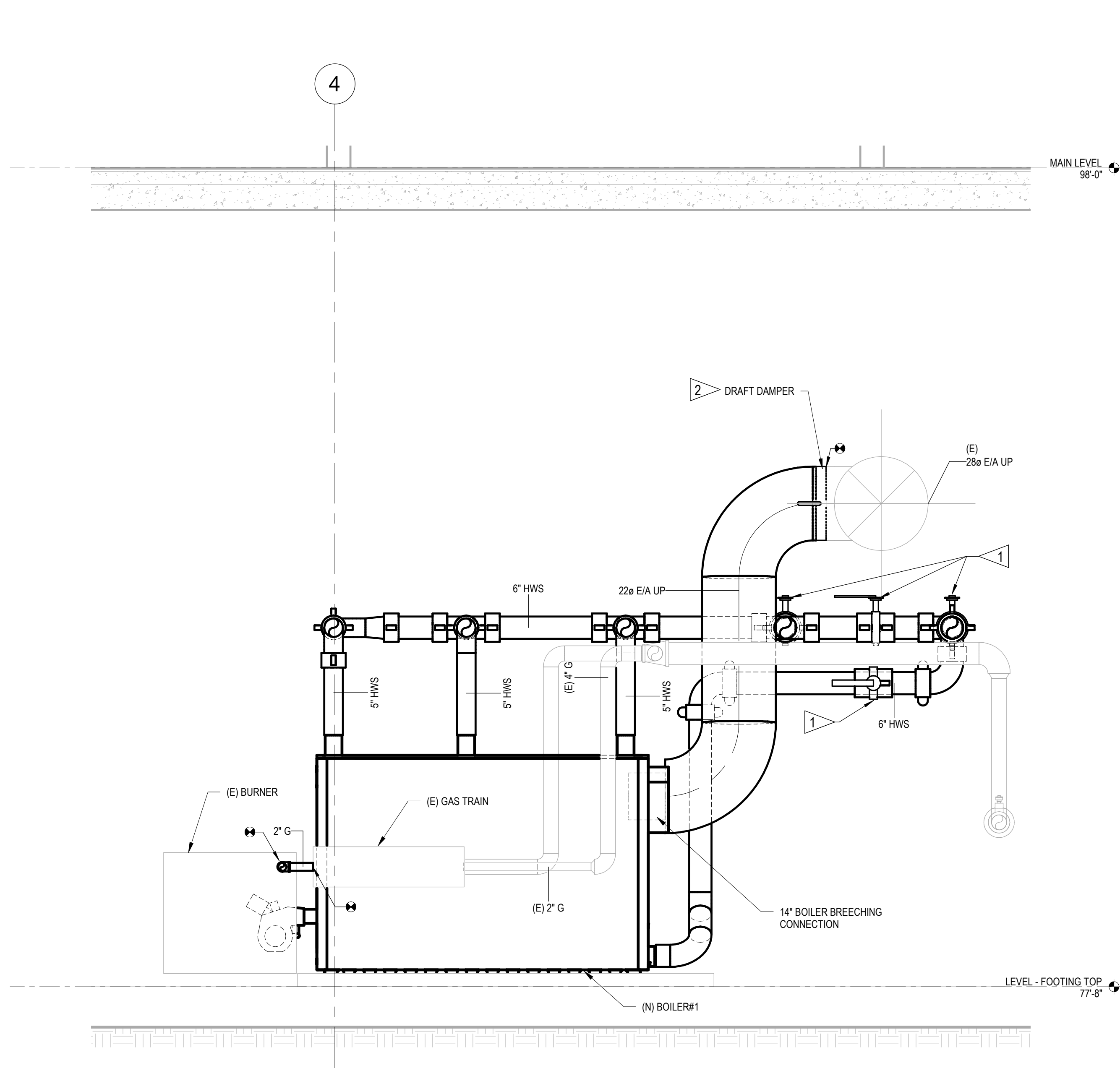
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SHEET NOTES

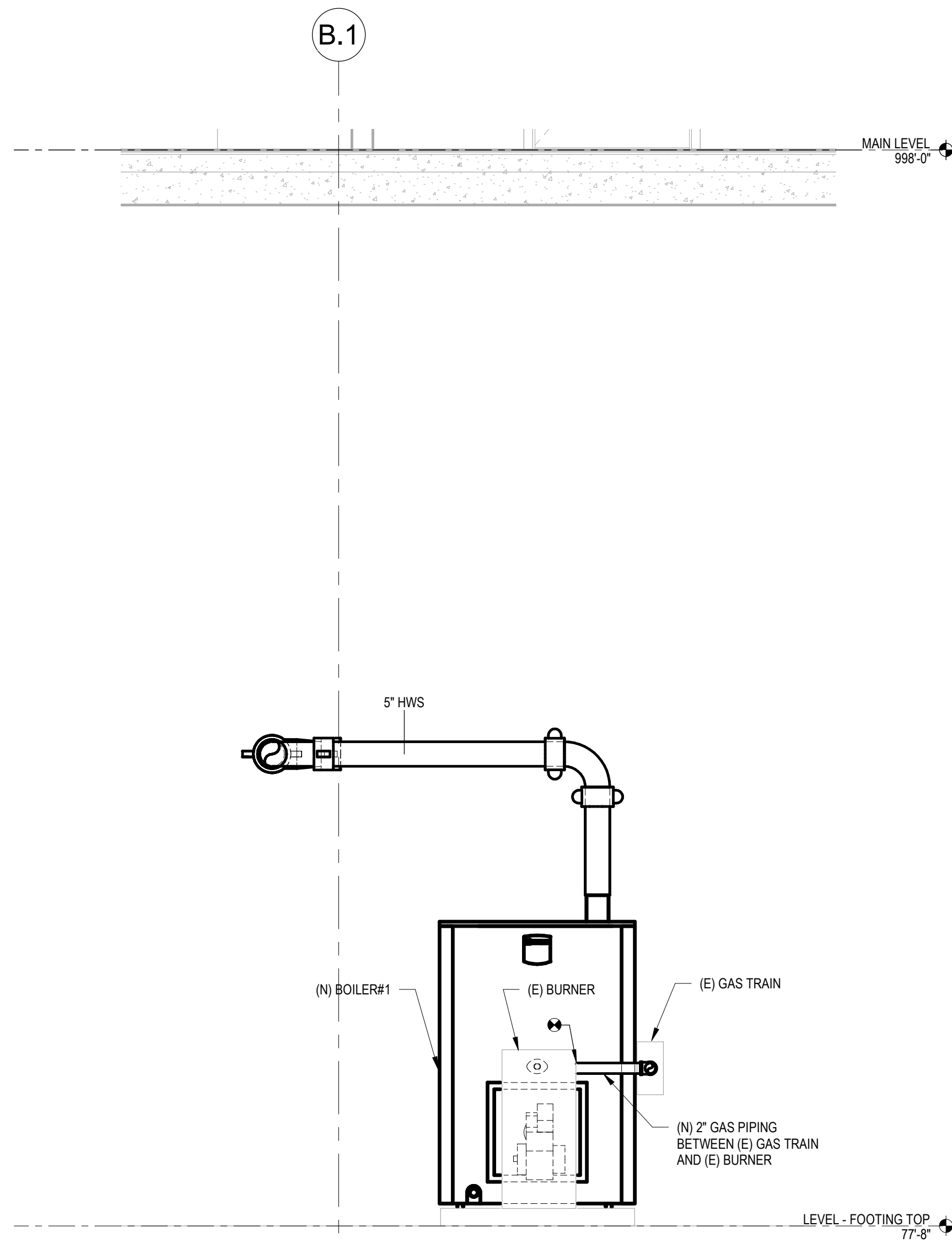
- 1 VALVE TO BE INSTALLED IN A BRIEF OUTAGE WHEN BOILER 2 CAN BE SHUT DOWN. BOILER 1 CAN THEN BE INSTALLED WITH BOILER 2 IN OPERATION.
- 2 INSTALL DRAFT DAMPER IN A BRIEF OUTAGE WHEN BOILER 2 CAN BE SHUT DOWN. BOILER 1 CAN THEN BE INSTALLED WITH BOILER 2 IN OPERATION.



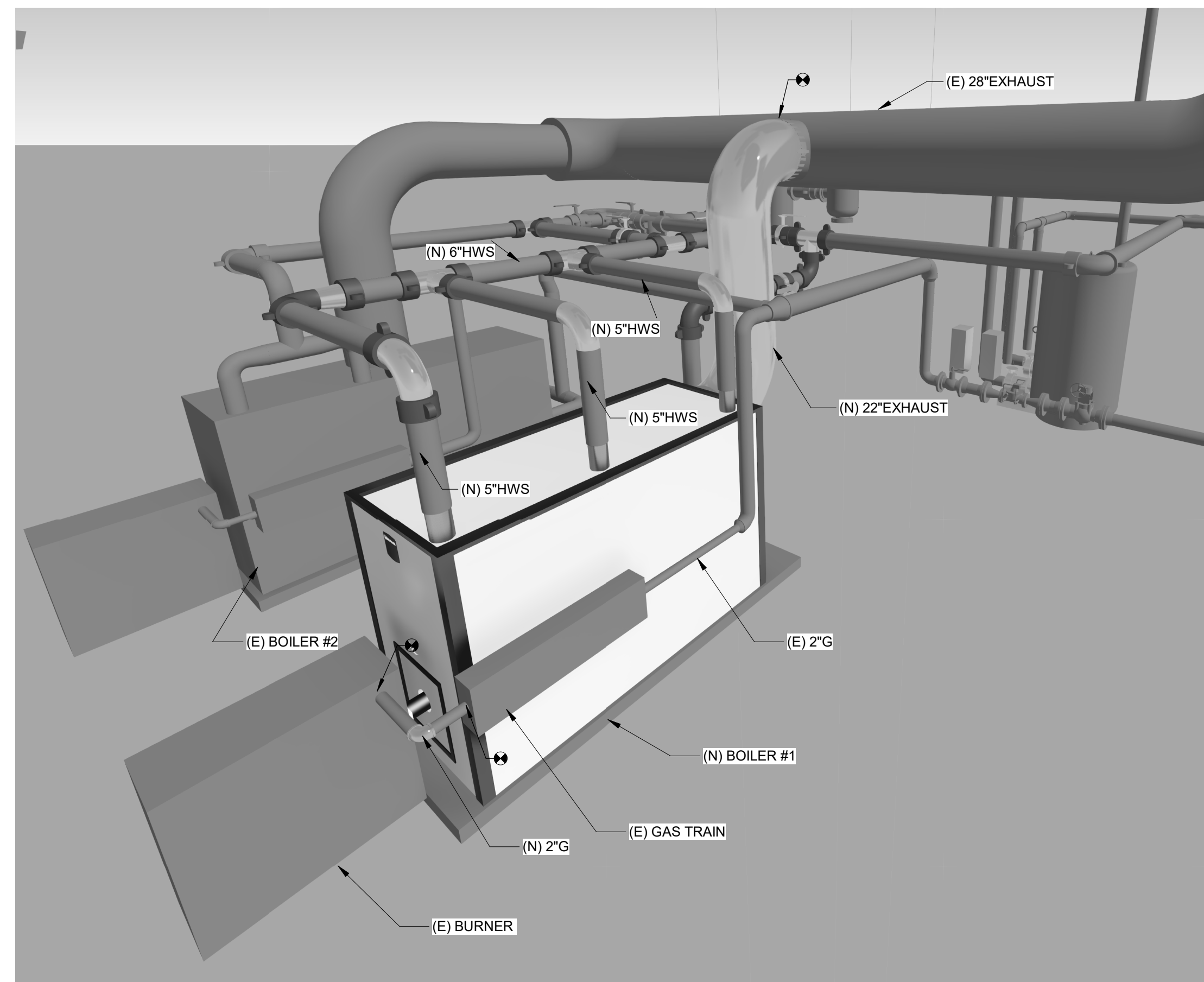
1
SECTION
1/2" = 1'-0"



2
SECTION
1/2" = 1'-0"



3
SECTION
1/2" = 1'-0"



4
ISOMETRIC VIEW



PLAZA COOLING TOWER & BOILER REPLACEMENT

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SEQUENCE OF
OPERATION

SHEET NO:

M-401

SHEET of

CONTROLS - SCOPE OF WORK

1. THE CONTROL DIAGRAMS AND DESCRIPTIONS OF OPERATION SHOWN ON THIS PLAN HAVE BEEN TAKEN FROM THE EXISTING CONTROLS DRAWING SET PREVIOUSLY DEVELOPED BY ATS INLAND NW. THE INTENT IS TO ALLOW FOR THE NEW CONTROLS CONTRACTOR TO GET FAMILIAR WITH THE EXISTING POINTS ASSOCIATED WITH EACH EQUIPMENT BEING REPLACED. THE DESCRIPTION OF OPERATION SHOWN ON THESE SHEETS IS THE EXISTING CONTROLS LOGIC THAT IS CURRENTLY IMPLEMENTED AND IS FOR REFERENCE ONLY.
2. THE CONTROLS CONTRACTOR SHALL INTEGRATE THE EXISTING POINTS SHOWN ON THESE DIAGRAMS TO THE NEW FLUID COOLER, CONDENSER WATER PUMP, HEATING BOILER, AND HEATING HOT WATER PUMPS. CONTROLS CONTRACTOR SHALL CHECK ALL POINTS DISPLAY AT THE USER FRONT END TO MATCH EXISTING.
3. CONTROLS CONTRACTOR SHALL SUBMIT SHOP DRAWINGS FOR ENGINEERING REVIEW.

DOAS AND CONDENSER WATER PLANT (E) SEQUENCE FOR REFERENCE

DOAS AND CONDENSER WATER PLANT SEQUENCE OF OPERATION.

1. GENERAL
 - A. CONDENSER WATER SYSTEM AND DOAS UNIT SERVE WATER-SOURCE HEAT PUMPS. REFER TO SEPARATE SEQUENCE OF OPERATION FOR ADDITIONAL INFORMATION.
 - B. PROVIDE OFFSETS, DEAD BANDS, TIME DELAYS AND SAFETIES FOR A FULLY FUNCTIONAL AND OPTIMIZED SYSTEM THAT LIMITS SHORT CYCLING.
2. CONDENSER WATER SYSTEM
 - A. THE CONDENSER WATER PUMP SHALL BE ENABLED WHEN ASSOCIATED HEAT PUMPS ARE OPERATING IN OCCUPIED MODE, OR WHEN ANY HEAT PUMP IS CALLING FOR HEATING OR COOLING IN OCCUPIED MODE.
 - B. THE CONDENSER WATER SYSTEM SHALL BE ENABLED FOR A MINIMUM OF 30 MINUTES TO PREVENT SHORT CYCLING OF EQUIPMENT.
 - C. IF THE CONDENSER WATER TEMPERATURE FALLS BELOW 60 DEGREES FAHRENHEIT, THE HEAT EXCHANGER CONTROL VALVE SHALL MODULATE TO MAINTAIN A MINIMUM 60 DEGREES FAHRENHEIT CONDENSER WATER TEMPERATURE.
 - D. IF THE CONDENSER WATER TEMPERATURE RISES ABOVE 80 DEGREES FAHRENHEIT, ENABLE THE COOLING TOWER IN THE FOLLOWING MANNER TO MAINTAIN A MAXIMUM 80 DEGREES FAHRENHEIT CONDENSER WATER TEMPERATURE:
 - a. FIRST STAGE: ENABLE BASIN PUMP AND COOLING TOWER FAN AT LOW SPEED.
 - b. SECOND STAGE: ENABLE BASIN PUMP AND COOLING TOWER FAN AT HIGH SPEED.
3. DIRECT OUTSIDE AIR SYSTEM (DOAS)
 - A. THE DOAS UNIT SHALL BE ENABLED ON THE SAME OCCUPIED OPERATING SCHEDULE AS THE ASSOCIATED WATER-SOURCE HEAT PUMPS.
 - B. THE HOT WATER CONTROL VALVE SHALL BE MODULATED TO PROVIDE 65 DEGREES FAHRENHEIT SUPPLY AIR TEMPERATURE.
 - C. THE DOAS UNIT SHALL BE DISABLED DURING MORNING WARM UP AND MORNING COOL DOWN MODE.
 - D. DOAS WAS LEFT DISABLED THROUGH DDC AT PROJECT CLOSEOUT. TAB CONTRACTOR DETERMINED THAT DOAS WAS NOT NEEDED TO PROVIDE MINIMUM AIRFLOW TO HP UNITS.
4. EXHAUST FANS
 - A. EF-6 AND EF-7 MONITOR STATUS ONLY.
5. ALARMING
 - A. "CONDENSER WATER SUPPLY TEMPERATURE OUTSIDE OF SET POINT": GENERATE AN ALARM WHEN THE CONDENSER WATER SUPPLY TEMPERATURES IS MORE THAN 5 DEGREES FAHRENHEIT ABOVE THE HIGH LIMIT OR 5 DEGREES FAHRENHEIT BELOW THE LOW LIMIT FOR AT LEAST 5 MINUTES.
 - B. "CONDENSER WATER PUMP FAILURE": GENERATE AN ALARM WHEN THE CONDENSER WATER PUMP STATUS DOES NOT MATCH THE COMMAND (CALL TO RUN FAIL).
 - C. "CHANGE FILTER": GENERATE AN ALARM WHEN THE DOAS UNIT FILTER SWITCH IS TRIPPED. AT TIME OF PROJECT, NO FILTER RACK INSTALLED ON THE DOAS. PRESSURE SWITCH IS INSTALLED FOR FUTURE.
 - D. "SUPPLY FAN FAILURE": GENERATE AN ALARM WHEN THE DOAS SUPPLY FAN STATUS DOES NOT MATCH THE COMMAND (CALL TO RUN FAIL).

HOT WATER PLANT (E) SEQUENCE FOR REFERENCE

HOT WATER PLANT SEQUENCE OF OPERATION.

1. GENERAL
 - A. THE HOT WATER SYSTEM SERVES 2-PIPE AND 4-PIPE FAN COILS, UNIT HEATERS, MULTI-ZONE UNITS, AND 4-PIPE ROOFTOP UNITS.
 - B. THE BUILDING AUTOMATION SYSTEM SHALL ENABLE AND STAGE THE BOILERS PER CONTROL CONTRACTOR PROVIDED SEQUENCE TO MAINTAIN THE ACTIVE LOOP HOT WATER TEMPERATURE SETPOINT.
 - C. PROVIDE OFFSETS, DEAD BANDS, TIME DELAYS AND SAFETIES FOR FULLY FUNCTIONAL AND OPTIMIZED SYSTEM THAT LIMITS SHORT CYCLING.
2. HOT WATER SYSTEM OPERATION
 - A. THE HOT WATER SYSTEM SHALL BE ENABLED WHENEVER OUTSIDE AIR TEMPERATURE IS LESS THAN 65 DEGREES FAHRENHEIT AND AT LEAST ONE HOT WATER VALVE IS OPEN TO ANY DEGREE.
 - B. HOT WATER SYSTEM TO BE ENABLED A MINIMUM OF 30 MINUTES TO PREVENT SHORT CYCLING OF EQUIPMENT.
 - C. BOILERS
 - a. THE BOILERS SHALL BE OPERATED IN A LEADLAG CONFIGURATION. THE LEAS BOILER SHALL CHANGE WEEKLY BASED ON RUNTIME.
 - b. THE BOILERS SHALL BE ENABLED AND STAGED TO MAINTAIN 180 DEGREES FAHRENHEIT HOT WATER SUPPLY TEMPERATURE.
 - c. THE COMBUSTION AIR DAMPERS SHALL BE OPEN WHENEVER ANY BOILER IS ENABLED.
3. ALARMING
 - A. "HOT WATER SUPPLY TEMPERATURE OUTSIDE OF SET POINT": GENERATE AN ALARM WHEN THE HEATING WATER SUPPLY TEMPERATURE IS NOT WITHIN 20 DEGREES FAHRENHEIT OF SET POINT. TIME DELAY SHALL BE OBSERVED AFTER A CHANGE IN COMMAND TO AVOID NUISANCE ALARMS.
 - B. "HOT WATER PUMP FAILURE": GENERATE AN ALARM WHEN A HEATING WATER PUMP STATUS DOES NOT MATCH THE COMMAND.
 - C. "BOILER ALARM" GENERATE AN ALARM WHEN ANY BOILER ALARM STATUS IS TRIPPED.

DOAS AND CONDENSER WATER PLANT (CT-2, HX-1, P-6) SYSTEM DIAGRAM

SCALE: NONE

SCOPE OF WORK - COOLING TOWER

REPLACE INSTRUMENTS AS NEEDED AND PROVIDE NEW WIRING TO INTEGRATE NEW EQUIPMENT TO EXISTING DDC CONTROLLER. REUSE EXISTING POINTS.

SCOPE OF WORK - CONDENSER WATER PUMP

REPLACE INSTRUMENTS AS NEEDED AND PROVIDE NEW WIRING TO INTEGRATE NEW EQUIPMENT TO EXISTING DDC CONTROLLER. REUSE EXISTING POINTS.

SCOPE OF WORK - HEATING HOT WATER PUMPS

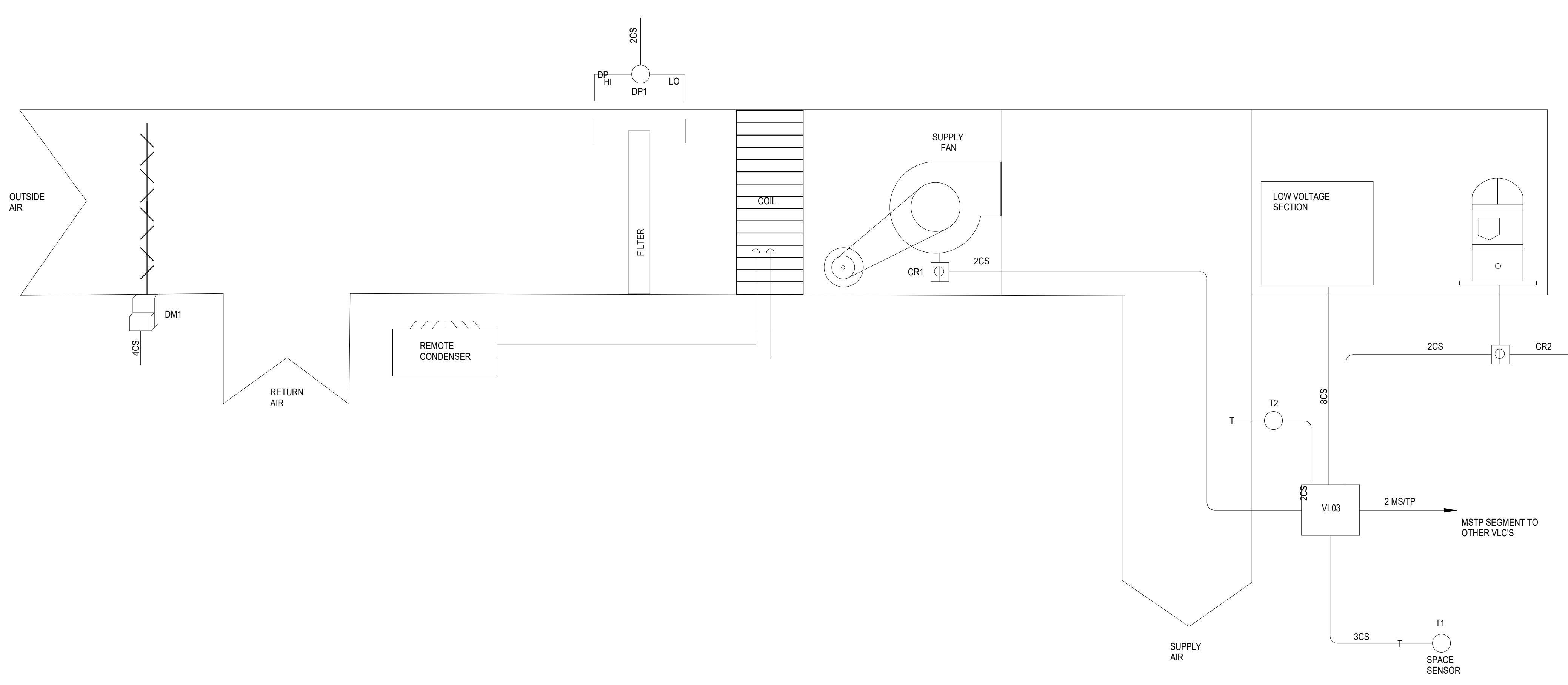
REPLACE INSTRUMENTS AS NEEDED AND PROVIDE NEW WIRING TO INTEGRATE NEW EQUIPMENT TO EXISTING DDC CONTROLLER. REUSE EXISTING POINTS.

SCOPE OF WORK - HOT WATER BOILER

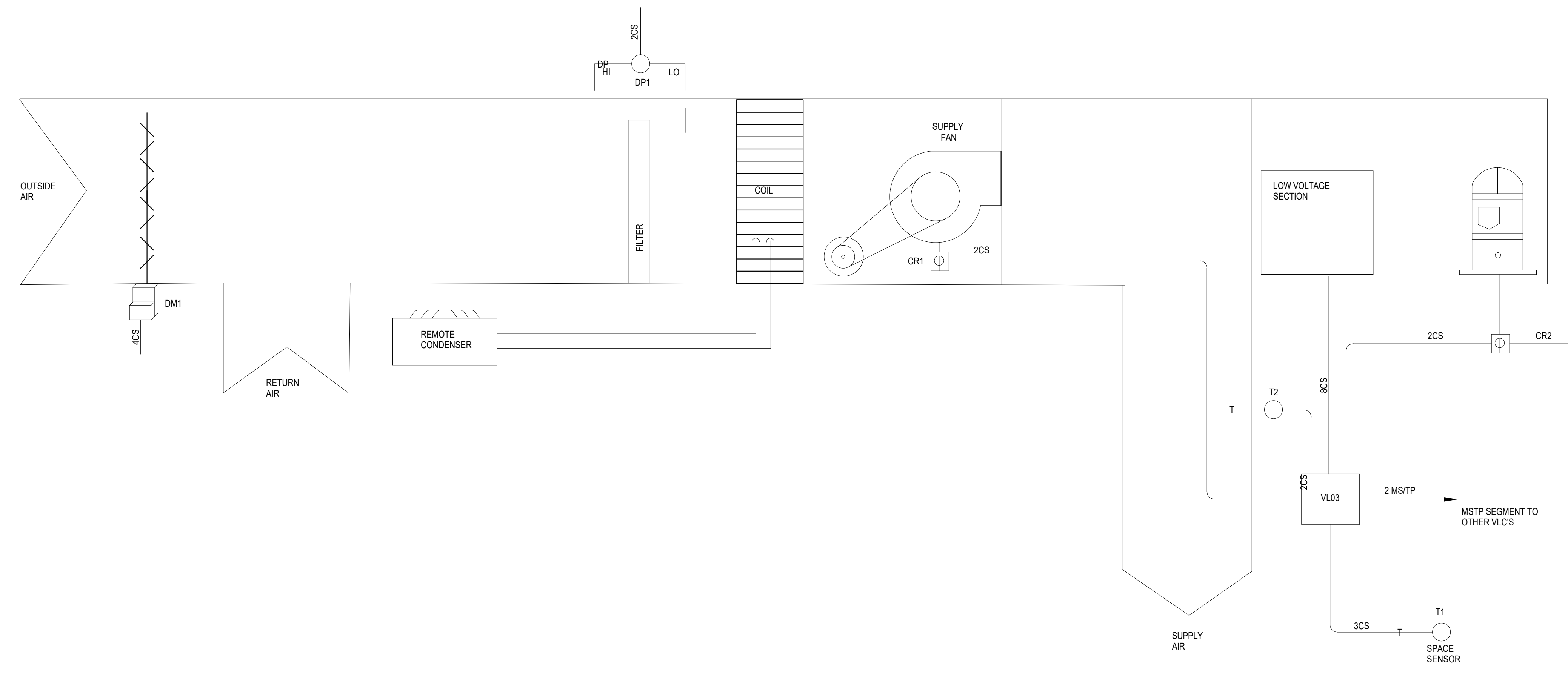
REPLACE INSTRUMENTS AS NEEDED AND PROVIDE NEW WIRING TO INTEGRATE NEW EQUIPMENT TO EXISTING DDC CONTROLLER. REUSE EXISTING POINTS.

HOT WATER PLANT SYSTEM DIAGRAM

SCALE: NONE



WATER-SOURCE HEAT PUMP HP-2 DIAGRAM
SCALE: NONE



WATER-SOURCE HEAT PUMP HP-3 THRU 6 DIAGRAM (TYP 4)
SCALE: NONE

CONTROLS - SCOPE OF WORK

THIS SHEET SHOWN FOR REFERENCE ONLY. NO WORK ON THIS EQUIPMENT.

WATER-SOURCE HEAT PUMP (E) SEQUENCE FOR REFERENCE

- WATER-SOURCE HEAT PUMP SEQUENCE OF OPERATION (HP-1 THRU HP-6).
- GENERAL
 - EACH PUMP OPERATES IN ANY ONE OF FOUR MODELS: MORNING WARM UP, MORNING COOL DOWN, OCCUPIED, OR UNOCCUPIED.
 - OCCUPIED AND UNOCCUPIED TIME OF DAY SCHEDULE FOR EACH HEAT PUMP IS PROVIDED.
 - OCCUPIED MODE
 - THE SUPPLY FAN OPERATES CONTINUOUSLY.
 - TEMPERATURE CONTROL
 - THE COMPRESSOR OPERATES IN COOLING OR HEATING MODE AS NECESSARY TO MAINTAIN OCCUPIED ZONE TEMPERATURE SETPOINTS.
 - ON A CALL FOR HEATING OR COOLING, A 30 SECOND TIME DELAY IS PROVIDED BEFORE COMPRESSOR ENABLEMENT TO ENSURE CONDENSER WATER PUMP FLOW IS ESTABLISHED.
 - DIRECT OUTSIDE AIR UNIT INTERLOCK
 - THE DOAS UNIT ASSOCIATED WITH THE HEAT PUMP OPERATES ON A STAND-ALONE SCHEDULE.
 - DAMPER CONTROL
 - HP-1: THE OUTSIDE AIR DAMPER IS FIXED POSITION AND WILL NOT BE CONTROLLED, VENTILATION TO BE PROVIDED WHEN ASSOCIATED DOAS IS ENABLED.
 - HP-2 THRU HP-6: THE OUTSIDE AIR DAMPER OPENS TO MINIMUM POSITION WHEN THE UNIT IS ENABLED IN OCCUPIED MODE.
 - UNOCCUPIED MODE
 - THE SUPPLY FAN IS OFF AND OUTSIDE AIR DAMPER CLOSED (IF APPLICABLE).
 - THE ASSOCIATED DOAS UNIT IS ENABLED OR DISABLED ON A SEPARATE SCHEDULE.
 - THE COMPRESSOR IS DISABLED.
 - IF THE ZONE TEMPERATURE FALLS BELOW THE UNOCCUPIED MODE HEATING SETPOINT, FOLLOW THE MORNING WARM UP SEQUENCE UNTIL ZONE TEMPERATURE IS 2 DEGREES FAHRENHEIT ABOVE UNOCCUPIED MODE HEATING SETPOINT.
 - IF THE ZONE TEMPERATURE RISES ABOVE THE UNOCCUPIED MODE COOLING SETPOINT, FOLLOW THE MORNING COOL DOWN MODE SEQUENCE UNTIL ZONE TEMPERATURE IS 2 DEGREES FAHRENHEIT BELOW UNOCCUPIED MODE COOLING SETPOINT.
 - MORNING WARM UP/ COOL DOWN MODE
 - AN OPTIMAL START PROGRAM IS PROVIDED THAT USE THE OUTSIDE AIR TEMPERATURE AND ZONE TEMPERATURE TO DETERMINE START TIME SO THAT THE OCCUPIED ZONE TEMPERATURE SETPOINT IS ACHIEVED 15 MINUTES PRIOR TO OCCUPANCY.
 - THE SUPPLY FAN RUNS CONTINUOUSLY.
 - THE COMPRESSOR IS ENABLED IN HEATING OR COOLING MODE AS NEEDED TO MEET OCCUPIED ZONE TEMPERATURE SETPOINT.
 - THE ASSOCIATED DOAS UNIT REMAIN DISABLED ACCORDING TO ITS OWN OPERATING SCHEDULE.
 - ALARMS
 - "LOW ZONE TEMPERATURE ALARM": GENERATE AN ALARM WHEN THE ZONE TEMPERATURE IS 5 DEGREES FAHRENHEIT BELOW SETPOINT FOR 15 MINUTES.
 - "HIGH ZONE TEMPERATURE ALARM": GENERATE AN ALARM WHEN THE ZONE TEMPERATURE IS 5 DEGREES FAHRENHEIT ABOVE SETPOINT FOR 15 MINUTES.
 - "SUPPLY FAN FAILURE": GENERATE AN ALARM WHEN THE SUPPLY FAN IS COMMANDED ON, BUT STATUS IS NOT PROVEN AFTER 30 SECONDS (CALL TO RUN FAIL).
 - "COMPRESSOR FAILURE": GENERATE AN ALARM WHEN THE COMPRESSOR IS COMMANDED ON, BUT THE STATUS IS NOT PROVEN AFTER 30 SECONDS (CALL TO RUN FAIL).
 - "CHANGE FILTER": GENERATE AN ALARM WHEN THE DIRTY FILTER SWITCH IS TRIPPED.



PLAZA COOLING
TOWER & BOILER
REPLACEMENT

701 W Riverside Ave

Spokane Transit
Authority

Permit Plan Set

REV	DATE	DESCRIPTION

PROJ. NO.	200465
DRAWN	CEI
CHECKED	CEI
DATE	07/06/2021

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SHEET TITLE:

SEQUENCE OF
OPERATION

SHEET NO:

M-402

SHEET of



PLAZA COOLING
TOWER & BOILER
REPLACEMENT

701 W Riverside Ave

Spokane Transit
Authority

Permit Plan Set

REV	DATE	DESCRIPTION

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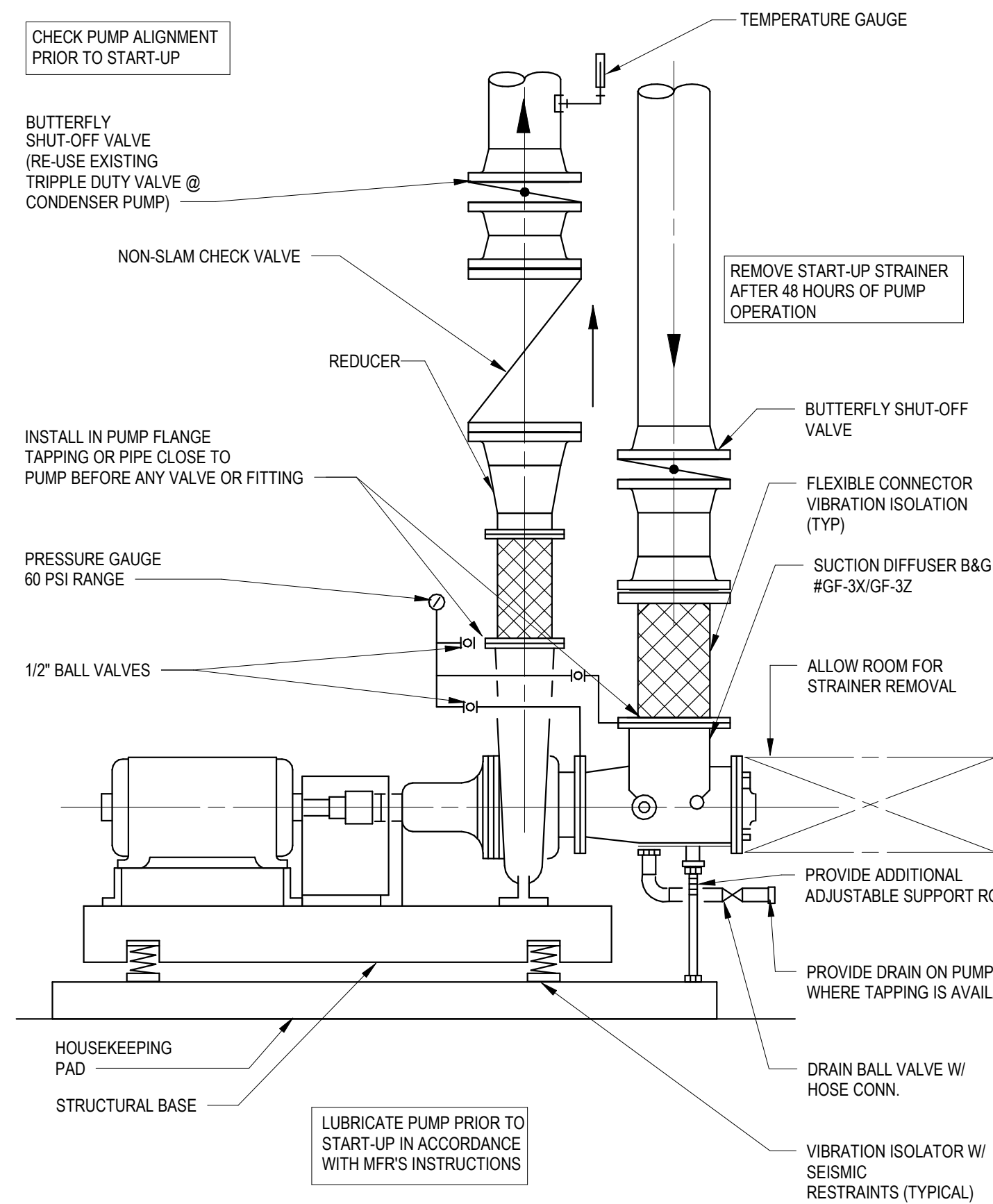
COFFMAN ENGINEERS

SHEET TITLE:
MECHANICAL
DETAILS

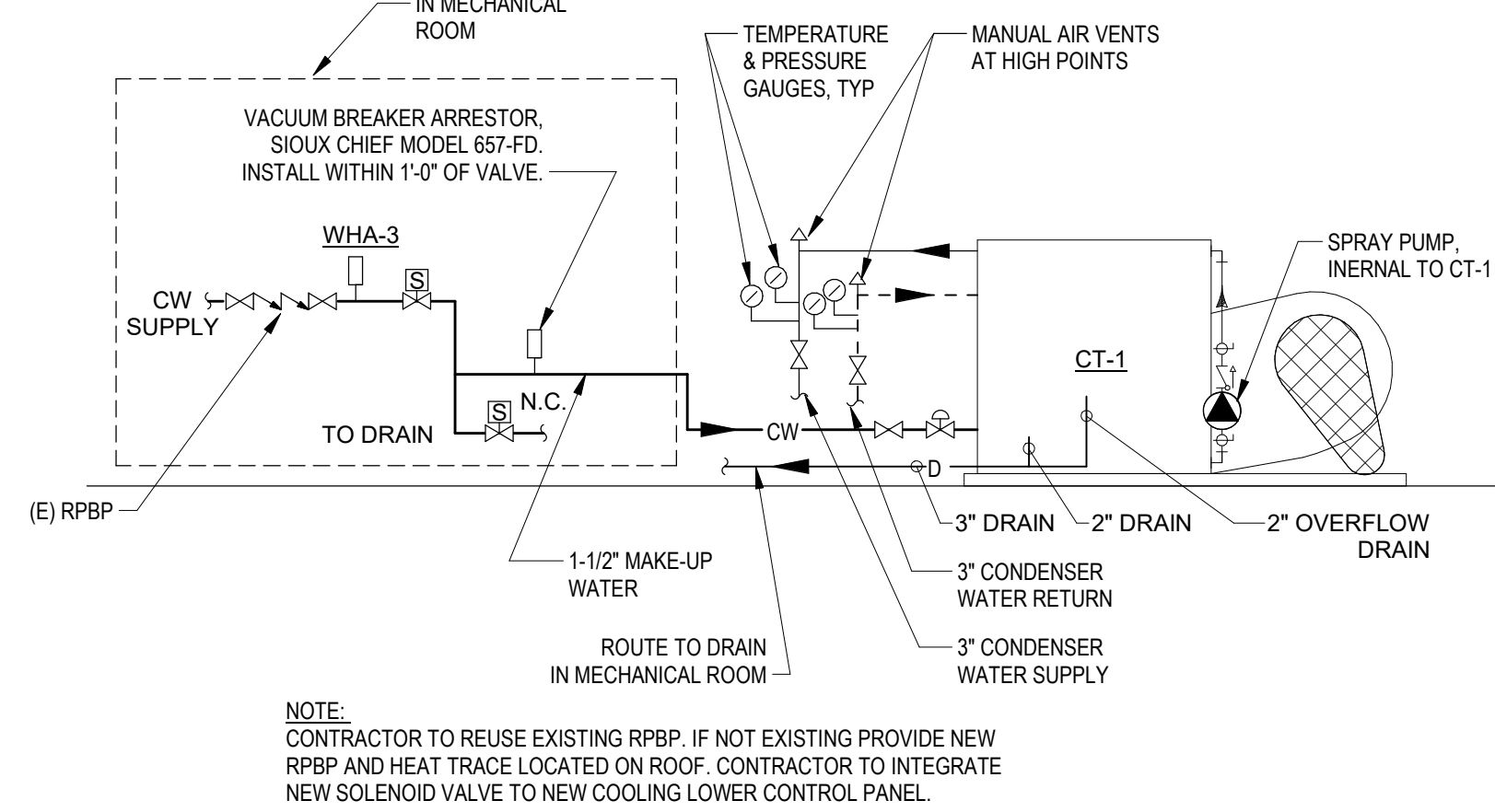
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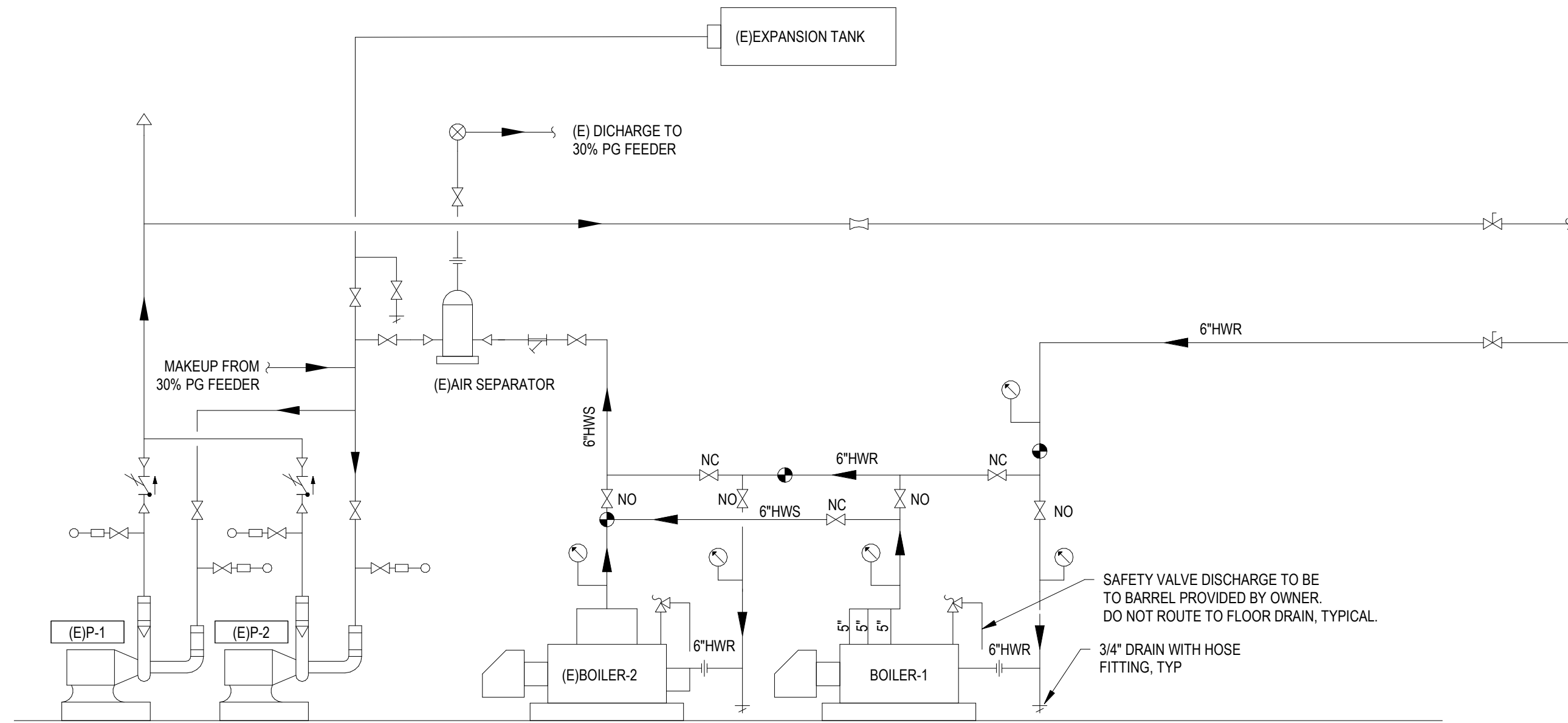
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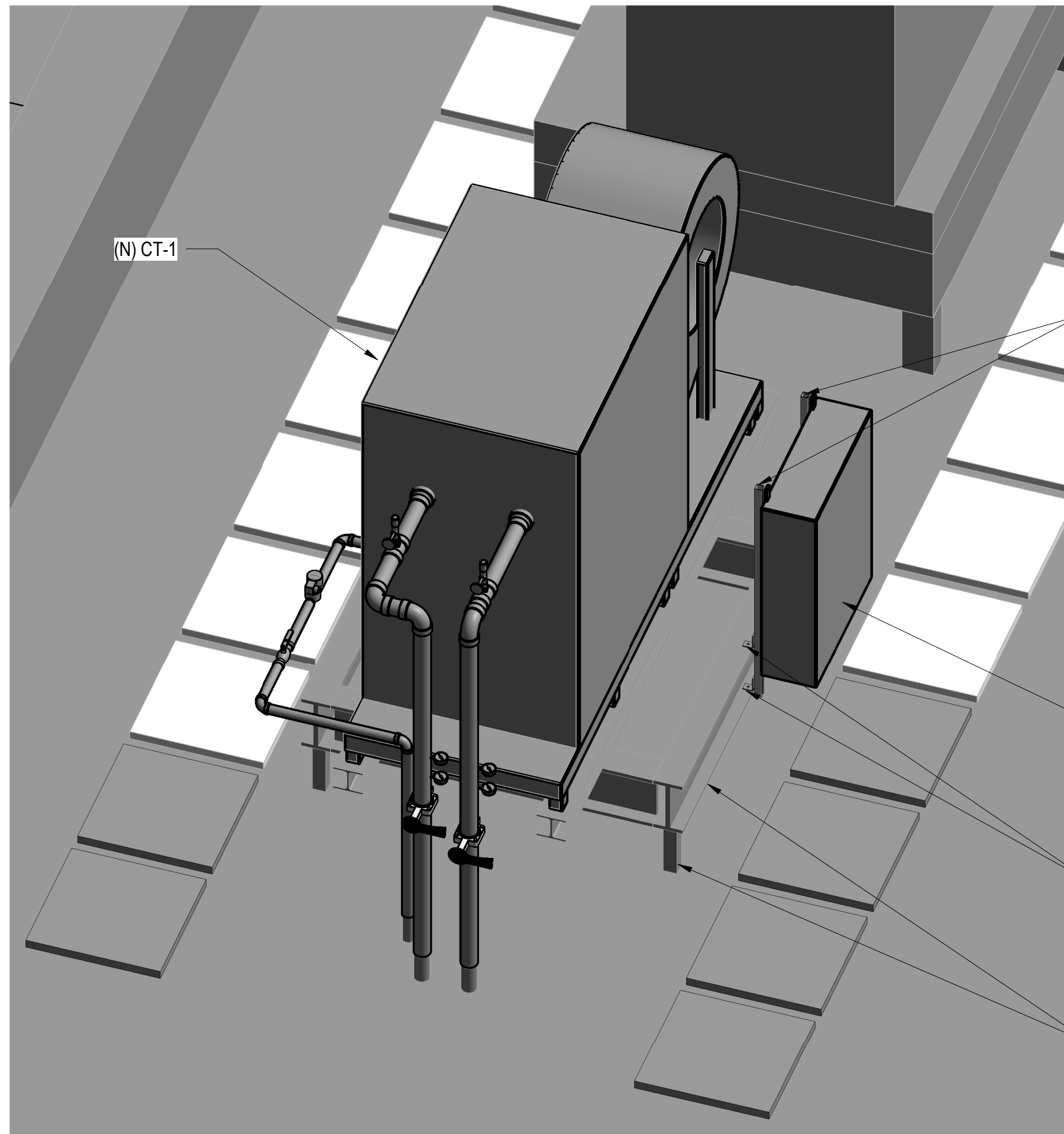
CONDENSER AND HEATING WATER PUMPS DETAIL



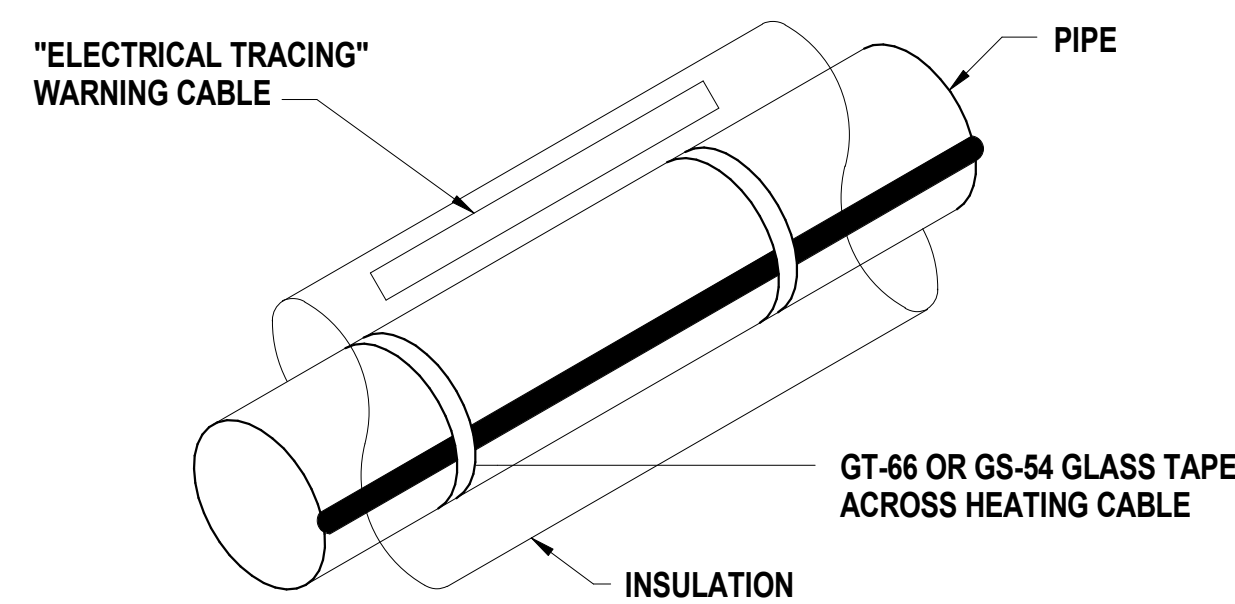
COOLING TOWER DETAIL



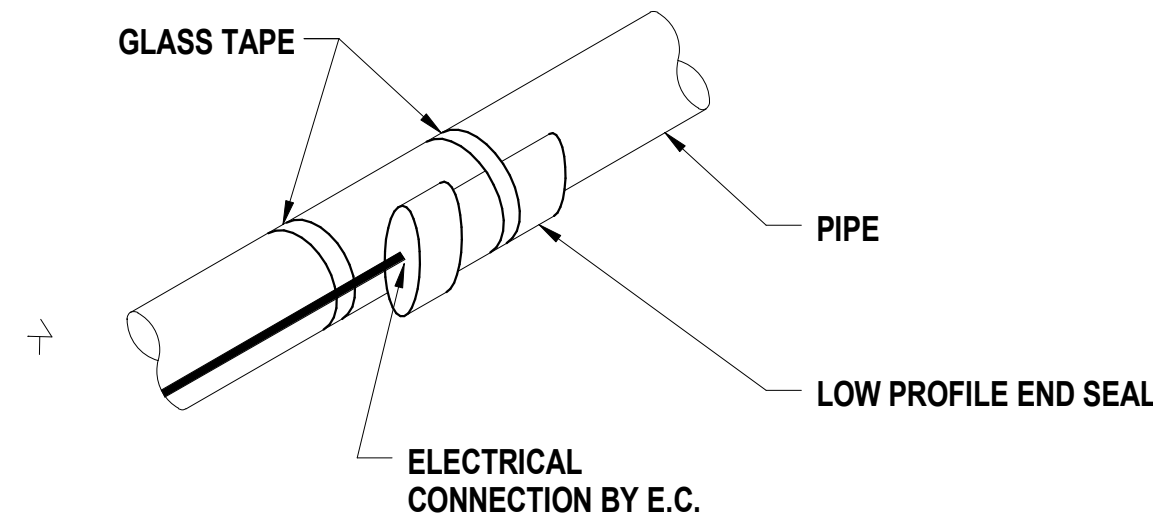
BOILER PIPING SCHEMATIC



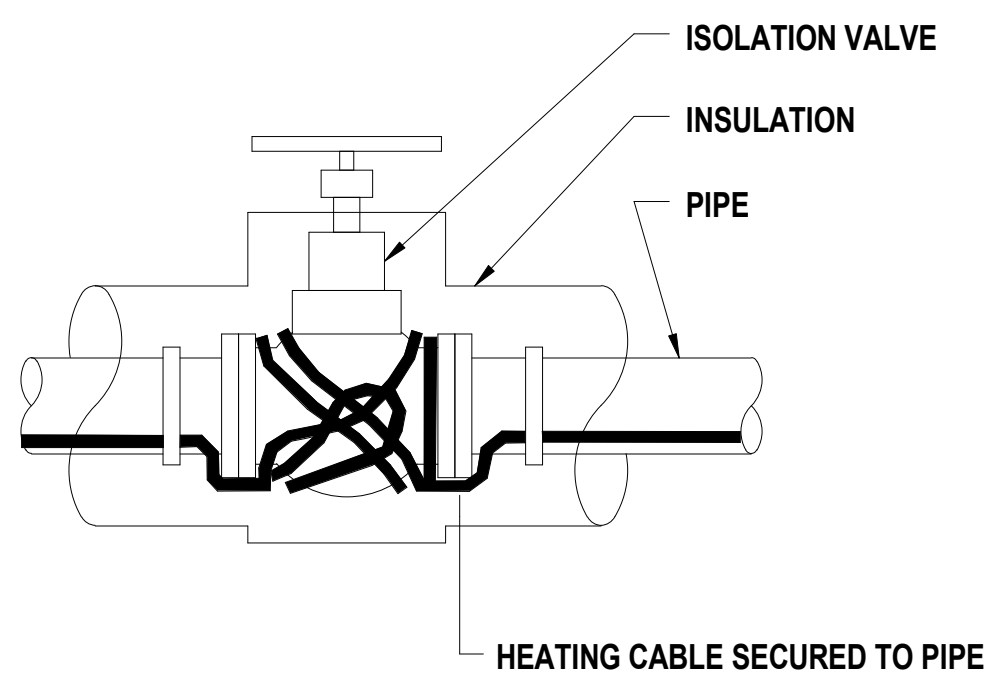
ISOMETRIC VIEW



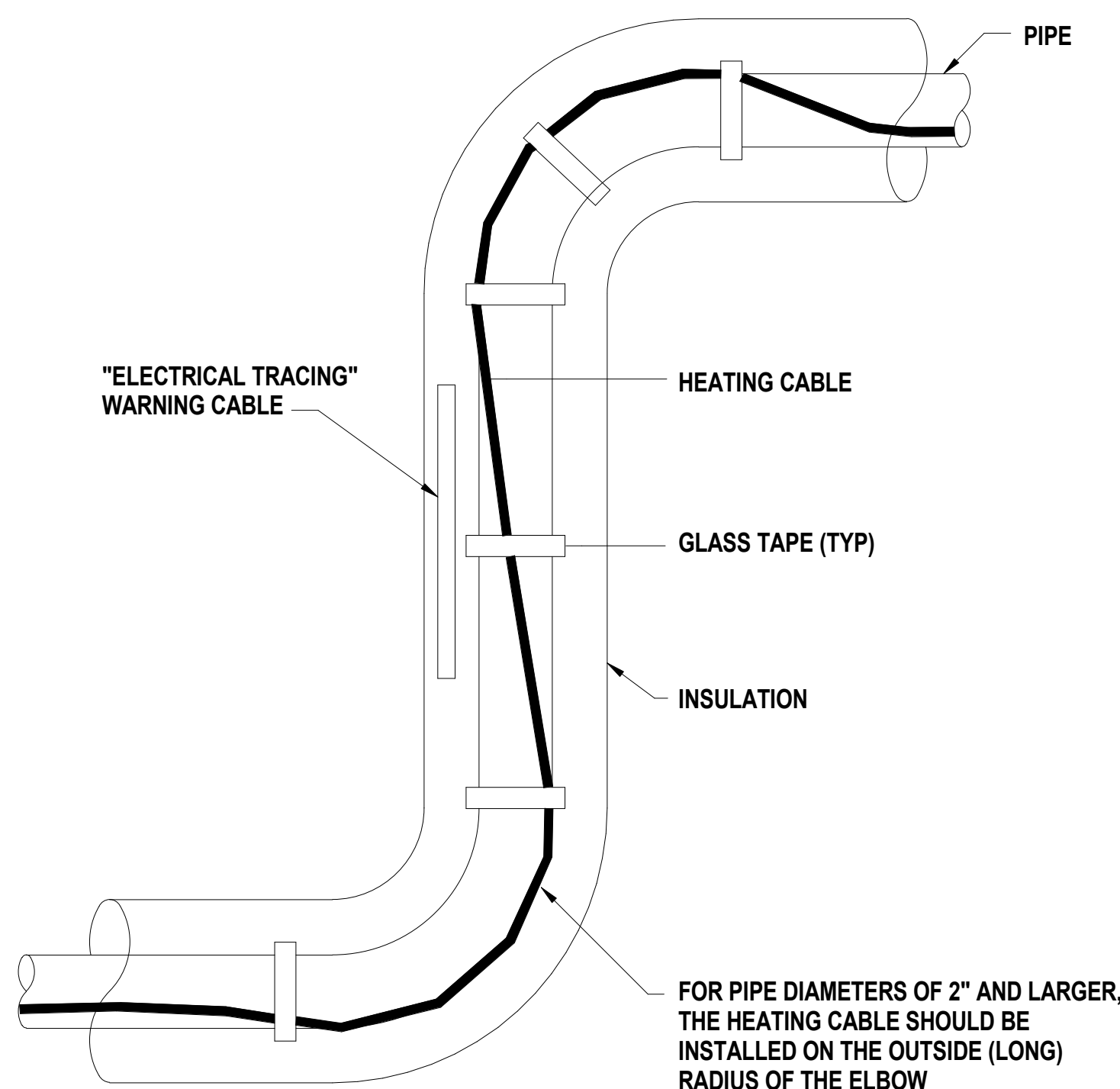
1 HEAT TRACE DETAIL - PIPE
M-502 SCALE: NTS



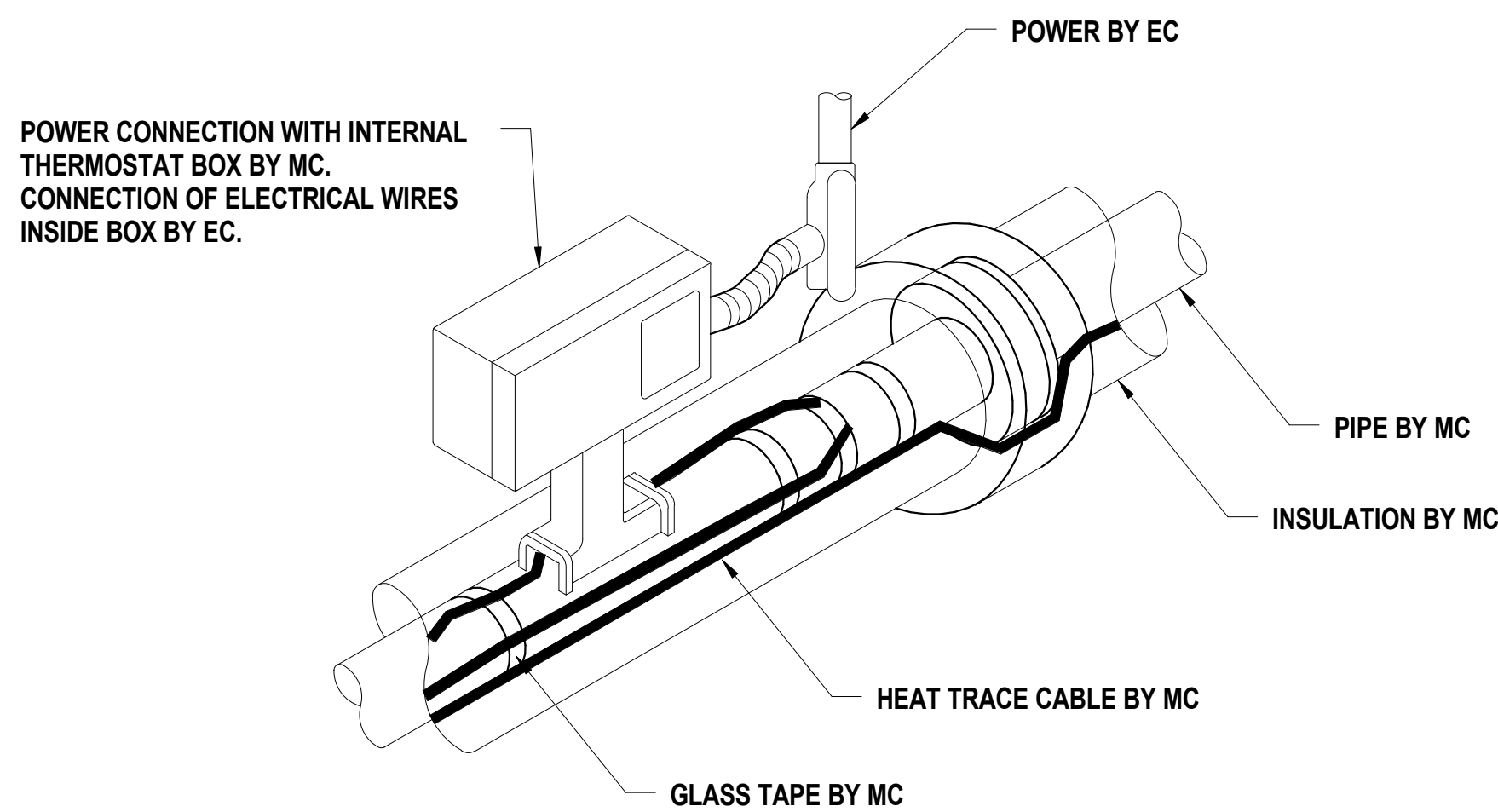
2 HEAT TRACE DETAIL - END SEAL
M-502 SCALE: NTS



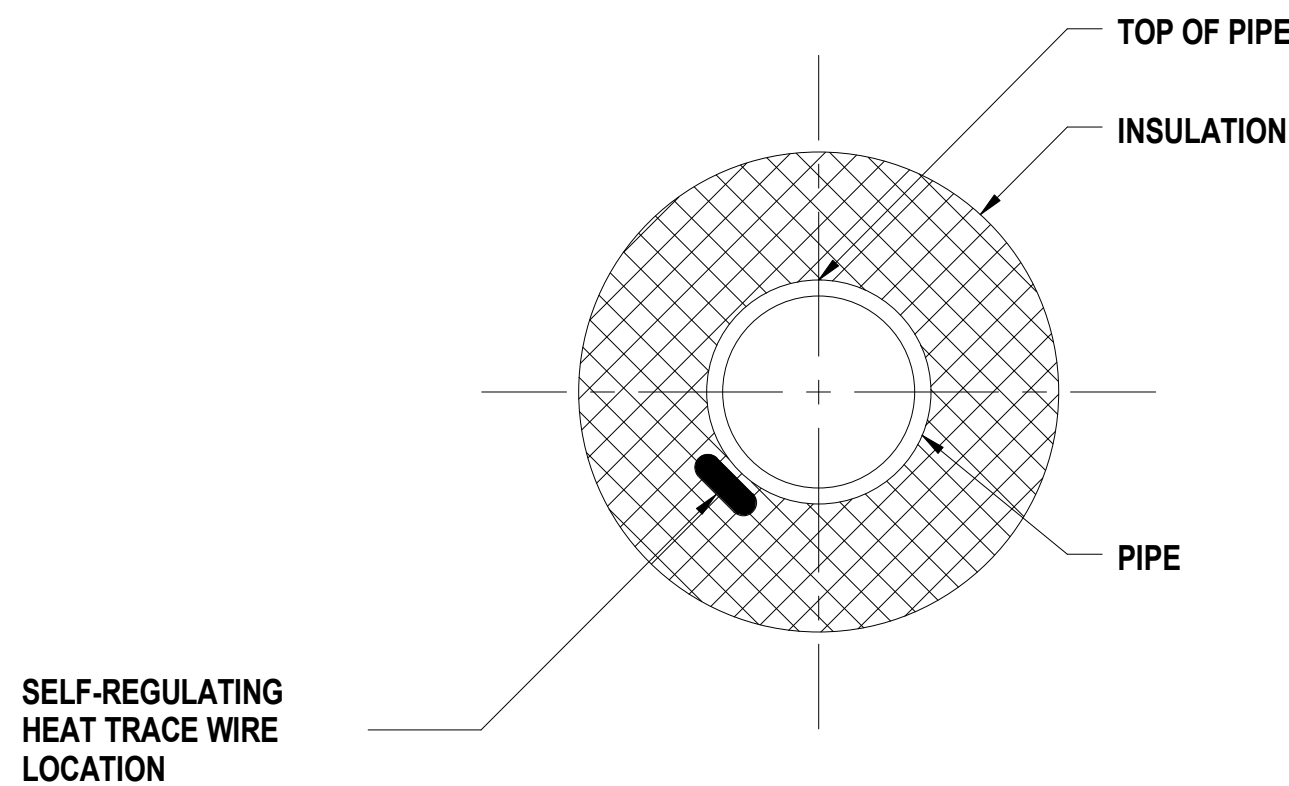
3 HEAT TRACE DETAIL - VALVE
M-502 SCALE: NTS



4 HEAT TRACE DETAIL - ELBOW
M-502 SCALE: NTS



5 HEAT TRACE DETAIL - POWER CONNECTION
M-502 SCALE: NTS



6 HEAT TRACE DETAIL - ONE WIRE INSTALLATION
M-502 SCALE: NTS



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SHEET TITLE:
HEAT TRACE
DETAILS

SHEET NO:

M-502

SHEET of